Firm Debt Portfolios and Factor Adjustment*

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Abstract

Using administrative firm-level data from Sweden, we document a robust asymmetry between debt maturity and firms' real adjustment: employment adjustment is substantially more closely associated with short-term debt, whereas capital adjustment is more closely associated with long-term debt. The same distinction appears in firms' longer-run evolution: as firms age and become more capital intensive, their debt portfolios shift toward longer maturities. We develop a dynamic heterogeneous-firm model in which employment, capital, and debt composition are jointly determined. Short-term debt provides a flexible source of within-period finance but is exposed to default risk and priced endogenously, while long-term debt is collateralized by capital, costly to adjust, and carried forward on the firm's balance sheet. These differences make short-term finance particularly responsive to current operating needs and long-term finance closely tied to capital and collateral capacity. The calibrated model reproduces the debt–factor asymmetry in both levels and adjustment rates, as well as the joint increase in capital intensity and long-term borrowing over the firm life cycle. A policy experiment that guarantees short-term creditor recovery shows that selectively making short-term finance cheaper can substantially reallocate firms' debt portfolios while lowering capital and output, highlighting that the maturity composition of corporate debt matters for the real effects of credit-support policies.

Keywords. Firm Dynamics; debt maturity; debt portfolios; employment adjustment; capital adjustment; financial frictions; default risk.

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1 Introduction

Firms do not simply choose how much to borrow; they also choose the maturity composition of their liabilities. These financing choices differ in their flexibility, collateral requirements, exposure to default, and implications for future balance sheets, and may therefore interact differently with firms' real decisions. Yet much of the empirical and quantitative literature either summarizes firms' liabilities through total debt or leverage, or studies debt maturity primarily through its implications for investment, refinancing risk, and default. This paper asks whether debt maturity is systematically related to the composition of firms' real adjustment. Using administrative firm-level data from Sweden, we show that it is: employment adjustment is much more closely associated with short-term debt adjustment, whereas capital adjustment is more closely associated with long-term debt adjustment. We refer to this pattern as the debt–factor asymmetry.

The asymmetry is remarkably stable across the sample and robust to alternative samples, additional balance-sheet controls, and specifications that absorb persistent firm heterogeneity and common time variation; it becomes especially pronounced around the Great Recession. The same distinction also appears in firms' longer-run evolution. As firms age, they become more capital intensive and increase the share of long-term debt in their liabilities. Moreover, after removing firm and time effects, periods in which a firm is more capital intensive relative to its own history are systematically associated with a larger long-term debt share. These findings suggest that debt maturity is connected not only to firms' overall financing conditions but also to the composition and persistence of their real adjustment: employment-intensive adjustment is associated more closely with short-term finance, while capital-intensive production structures are associated more closely with long-term finance.

To interpret these facts, we develop a dynamic heterogeneous-firm model in which employment, capital, and the composition of the debt portfolio are chosen jointly. The two debt instruments differ in the margins that are central to firms' real adjustment. Short-term debt provides flexible within-period finance but is exposed to default risk and priced competitively according to expected repayment and recovery. Long-term debt is collateralized by capital, priced at the risk-free rate, costly to adjust, and carried forward on the firm's balance sheet. Greater long-term borrowing therefore reduces the firm's current reliance on risky short-term credit, but it also uses collateral and reduces the residual liquidation value available to short-term creditors. The optimal debt portfolio consequently reflects a dynamic trade-off between current financing needs and future exposure to financial distress.

The mechanism rests on a dynamic trade-off between the two sources of finance. Greater long-term borrowing provides resources today and reduces the firm's current reliance on risky short-term credit. But long-term debt remains on the balance sheet. Because it is collateralized and has priority over short-term claims, a larger long-term position also reduces the residual assets available to short-term creditors and affects the terms on which the firm can obtain short-term finance in the future. Greater reliance on short-term borrowing has the opposite profile: it increases the firm's current exposure to repayment risk and adverse revenue realizations, but leaves less long-term debt to be carried forward. The optimal debt portfolio therefore balances current and future exposure to financial distress.

The price of short-term credit is itself part of this portfolio problem. Competitive lenders price short-term debt according to expected repayment and recovery. Borrowing terms therefore depend on the firm's productivity, capital, inherited long-term debt, and its current real and financial decisions. Changes in employment, capital, and long-term borrowing affect not only the amount of short-term finance required but also the probability of default and the value available to creditors when default occurs. The model consequently generates heterogeneity

in short-term borrowing costs and default probabilities across firms as an endogenous outcome of their production and financing decisions.

This feature connects the model to recent quantitative work with endogenous defaultable corporate debt. Arellano et al. [2019], for example, generate state-dependent borrowing terms as firms' production and financing decisions interact with default risk. Our setting adds an endogenous short–long debt portfolio to this environment. At the same time, the model builds on the maturity-choice mechanism in Jungherr and Schott [2021], where firms optimally trade off short- and long-term borrowing and debt maturity interacts with investment and default. The portfolio interaction is different in our environment because collateralized long-term claims affect the recovery position supporting risky short-term finance. This creates a direct link between the composition of the firm's liabilities, the price of short-term credit, and its real decisions.

The two factors of production interact differently with this financing problem. Employment responds relatively flexibly to current operating conditions and changes the scale of expenditures that must be financed before revenues are known. Capital is more persistent and, in addition to its productive role, changes the firm's collateral capacity, liquidation value, and ability to support long-term borrowing. As a result, changes in the relative costs and benefits of short- and long-term finance have different implications depending on whether the firm is adjusting employment or capital. The observed short-debt–employment and long-debt–capital relationships thus arise jointly from firms' optimal production and portfolio choices.

Our contribution is to connect firms' debt-maturity choices to the composition of their real adjustment. A large literature has established that financial conditions matter for both employment and investment Sharpe [1994], Chodorow-Reich [2014], Bacchetta et al. [2019], Ottonello and Winberry [2020], Cloyne et al. [2023], while another has studied why firms choose different debt maturities and how those choices affect refinancing risk, investment, and default Barclay and Smith [1995], Jungherr and Schott [2021]. We bring these margins together by studying employment and capital within the same financing environment. We show that the distinction between short- and long-term debt maps systematically into the distinction between the two factors of production: employment adjustment is more closely associated with short-term borrowing, whereas capital adjustment is more closely associated with long-term borrowing. This factor-specific dimension of firms' financing behavior is obscured when liabilities are summarized by total debt or leverage.

The evidence also extends the debt-maturity perspective beyond contemporaneous factor adjustment. Firms become more capital intensive and shift toward long-term borrowing as they age, and the relationship between capital intensity and debt composition remains visible within firms after removing persistent firm heterogeneity and common time variation. Thus, the connection between production structure and debt maturity is not simply a cross-sectional comparison between permanently different types of firms. Changes in firms' production structures are accompanied by systematic changes in the maturity composition of their liabilities.

The model provides a unified account of these facts by combining working-capital financing, collateralized borrowing, endogenous defaultable debt prices, and endogenous maturity choice. In this respect, the framework builds on the working-capital timing emphasized by Jermann and Quadrini [2012], Mendoza [2010], the role of productive assets as collateral in Kiyotaki and Moore [1997], Rampini and Viswanathan [2013], endogenous risky borrowing terms as in Arellano et al. [2019], and endogenous debt maturity as in Jungherr and Schott [2021]. The distinctive interaction in our environment is that long-term borrowing not only reduces the firm's current reliance on risky short-term finance, but also changes the recovery position of short-term creditors in future periods. Firms therefore trade off greater current exposure to short-term repayment risk against the future financing consequences of carrying long-term debt.

The same forces generate firm-life-cycle dynamics. Entrants begin without an inherited long-term debt position. As surviving firms grow and accumulate capital, their collateral capacity expands and they increase their use of long-term finance. At the same time, installed capital is partially irreversible and long-term debt is costly to adjust, so both evolve gradually. Employment and short-term borrowing can respond more rapidly to current conditions. The model therefore generates a joint evolution of firms' production and financing structures in which capital intensity and the long-term debt share rise with firm age, linking the analysis to the broader literature on finance and firm dynamics Cooley and Quadrini [2001], Clementi and Hopenhayn [2006].

Quantitatively, the calibrated model reproduces the main empirical patterns even though the debt–factor relationships and the life-cycle relationships are not targeted in the calibration. It generates a stronger association of short-term debt with employment and of long-term debt with capital, both in levels and in year-to-year adjustment. It also reproduces the increase in capital intensity and in the long-term debt share over the firm life cycle, together with the positive relationship between capital intensity and long-term borrowing. These results show that the same portfolio mechanism can account for both the factor-specific financing patterns and the longer-run joint evolution of firms' real and financial structures.

We use the model to study a government guarantee that increases short-term creditor recovery in default. The guarantee lowers the price of short-term finance and induces firms to rely more heavily on short-term borrowing. At small interventions, the improvement in financing conditions also supports an expansion in long-term borrowing, but as the guarantee becomes larger firms increasingly tilt their portfolios toward the supported short-term instrument. In equilibrium, the policy raises the wage, reduces capital accumulation and output, and increases firm exit. At the same time, the debt–factor asymmetry remains intact and becomes stronger along several dimensions. The experiment illustrates that selectively reducing the cost of one maturity of corporate debt can have effects that differ sharply from a uniform relaxation of financing constraints: changes in the composition of finance feed back into firms' real decisions.

The remainder of the paper proceeds as follows. Section 2 presents the empirical evidence on debt portfolios, factor adjustment, firm life-cycle patterns, and capital intensity. Section 3 develops the dynamic heterogeneous-firm model with endogenous debt-portfolio choice. Section 4 characterizes the forces governing firms' short- and long-term financing decisions and their interaction with employment, capital, default risk, and short-term credit terms. Section 5 describes the calibration of the model. Section 6 evaluates the quantitative model against the debt–factor asymmetry, the life-cycle evolution of debt composition and capital intensity, and the relationship between capital intensity and long-term borrowing. Section 7 studies the government guarantee to short-term creditors and its implications for debt composition and real activity. Section 8 concludes.

Literature review. Our paper is related to a large literature studying how firms' financial positions shape real activity. Early work emphasized the role of balance-sheet strength and financing constraints for firms' responses to aggregate shocks Gertler and Gilchrist [1994], Bernanke et al. [1996, 1999]. On the investment side, a long literature has studied how access to external finance affects capital accumulation Fazzari et al. [1988], Whited [1992], Kaplan and Zingales [1997], while more recent work has emphasized heterogeneity in investment responses across firms with different financial positions Ottonello and Winberry [2020], Cloyne et al. [2023]. A parallel literature shows that financial conditions also matter importantly for employment. Sharpe [1994] relates leverage to employment cyclicality, while Chodorow-Reich [2014], Giroud and Mueller [2017], Benmelech et al. [2019] document large employment responses to credit disruptions and weak balance sheets. Bacchetta et al. [2019] emphasizes the role of liquid resources for employment, and Amberg et al. [2023] studies the state-dependent

provision of liquidity through credit lines. Closest to our emphasis on the composition of real inputs, ElFayoumi [2024] studies how financing conditions affect the relative demand for labor and capital. Our contribution differs in focusing on the maturity composition of debt itself. Rather than asking whether finance affects employment or investment, we show that short- and long-term debt are systematically associated with different margins of firms' real adjustment.

Our evidence also contributes to the corporate-finance literature on debt maturity. Classic theories emphasize debt overhang, liquidity risk, and the trade-off between short- and long-term claims Myers [1977], Diamond [1991]. Empirically, debt maturity varies systematically with firm characteristics, asset structure, risk, and borrowing capacity Barclay and Smith [1995], Stohs and Mauer [1996], Custódio et al. [2013]. Several papers are especially relevant to the capital side of our evidence. Norden and van Kampen [2013] documents a strong connection between tangible assets and long-term leverage, while Geelen et al. [2024] shows that firms match debt maturity to the economic life of their assets. At the same time, short-term debt need not be tied only to short-lived uses of funds: Kahl et al. [2015] shows that commercial paper can provide bridge financing for capital investment. Other work shows that maturity affects firms' exposure to refinancing conditions and can have real consequences during financial disruptions Almeida et al. [2012]. More recent quantitative work makes debt maturity an endogenous firm choice and studies its interaction with investment, leverage, default, and aggregate transmission Jungherr and Schott [2021], Jungherr et al. [2024]. Relative to this literature, our empirical contribution is to bring the employment margin directly into the study of debt maturity and to compare employment and capital within the same firm-level environment. The resulting debt–factor asymmetry provides a factor-specific dimension of maturity choice: employment adjustment is much more closely associated with short-term borrowing, whereas capital adjustment is more closely associated with long-term borrowing. Moreover, this distinction extends beyond contemporaneous adjustment. Firms become more capital intensive and rely relatively more on long-term debt as they mature, and the relationship between capital intensity and debt composition remains visible within firms after removing persistent firm heterogeneity and common time variation.

A related quantitative literature makes firms' financial positions endogenous state variables that evolve jointly with firm size, investment, and survival Cooley and Quadrini [2001], Clementi and Hopenhayn [2006], Midrigan and Xu [2014]. Our model draws on several mechanisms developed in this literature. Working-capital models emphasize that firms must finance expenditures before the revenues generated by production are realized Fuerst [1992], Neumeyer and Perri [2005], Christiano et al. [2005], Ravenna and Walsh [2006], Mendoza [2010], Jermann and Quadrini [2012]. Models of collateral constraints emphasize the dual role of productive capital as both an input into production and an asset that supports external finance Kiyotaki and Moore [1997], Rampini and Viswanathan [2010, 2013]. Quantitative models with endogenous default make borrowing terms themselves state dependent because firms' production and financing choices affect both default probabilities and creditor recovery Arellano et al. [2019], Khan et al. [2017]. Capital adjustment frictions, in turn, make firms' productive structure persistent over time Khan and Thomas [2008], Bloom [2009].

Our framework brings these mechanisms together through an endogenous short–long debt portfolio. Short-term debt finances within-period needs and is exposed to idiosyncratic revenue risk, while long-term debt is collateralized by capital and carried forward on the firm's balance sheet. Because long-term creditors have priority, the long-term debt position also changes the residual recovery value available to short-term creditors. Firms' choices of employment, capital, and debt composition therefore jointly determine current financing needs, default exposure, creditor recovery, and future financing conditions. This structure allows the model to speak simultaneously to the short-debt–employment and long-debt–capital relationships, to the evolution of debt composition

and capital intensity over the firm life cycle, and to the within-firm relationship between production structure and debt maturity.

Taken together, our contribution is therefore not simply to add another determinant of debt maturity or another financial friction to a firm-dynamics model. We document a new factor-specific organization of firms' debt portfolios, show that it appears both in short-run adjustment and in firms' longer-run evolution, and develop a quantitative mechanism in which the intertemporal trade-off between short- and long-term finance generates these patterns jointly. The fact that the calibrated model reproduces these relationships without targeting them provides an independent test of the mechanism and makes the maturity composition of debt relevant for evaluating policies that selectively alter the cost of one financing margin.

2 Empirical Evidence

This section documents a set of empirical regularities linking firms' debt portfolios to their employment and capital decisions. The main finding is a sharp asymmetry across the two components of debt. Changes in employment are much more closely associated with changes in short-term debt, whereas changes in capital are more closely associated with changes in long-term debt. These differences become particularly pronounced around the Great Recession. We also document a life-cycle pattern in which firms become more capital intensive and rely relatively more on long-term debt as they age.

Our objective in this section is descriptive. We do not interpret the relationships below as causal effects of debt on factor demand. Instead, we use them as empirical moments that any model of firms' financing and production decisions should be able to reproduce. This distinction is important because firms jointly determine their financing and production decisions in response to underlying shocks. The quantitative model developed below is designed to provide an economic mechanism that can account for these joint movements.

2.1 Data and Measurement

We use annual firm-level accounting data from the Swedish *Företagens ekonomi* (FE) database over the period 1997–2013. The data contain approximately 1.5 million firm-year observations covering about 245,000 firms. The accounts allow us to observe firms' employment, capital, sales, and balance-sheet positions, including short- and long-term debt.

Our main empirical analysis focuses on changes rather than levels. For a variable x_{it} belonging to firm i in year t, we measure its symmetric growth rate as

$$g_{it}^{x} \equiv \frac{x_{it} - x_{i,t-1}}{\frac{1}{2}\left(x_{it} + x_{i,t-1}\right)}. \tag{1}$$

The transformation in equation (1) treats expansions and contractions symmetrically and, unlike log differences, can accommodate observations in which a variable moves into or out of zero. This is particularly useful for debt variables because firms may enter or leave one part of the debt market over time.

Throughout the section, we use n_{it} and k_{it} to denote employment and capital, respectively, and s_{it} and b_{it} to denote short- and long-term debt. This notation anticipates the notation used in the model.

2.2 Empirical Specification

We begin by asking how changes in each component of the debt portfolio are associated with changes in employment and capital. For each outcome $y_{it} \in \{n_{it}, k_{it}\}$, our baseline specification can be written as

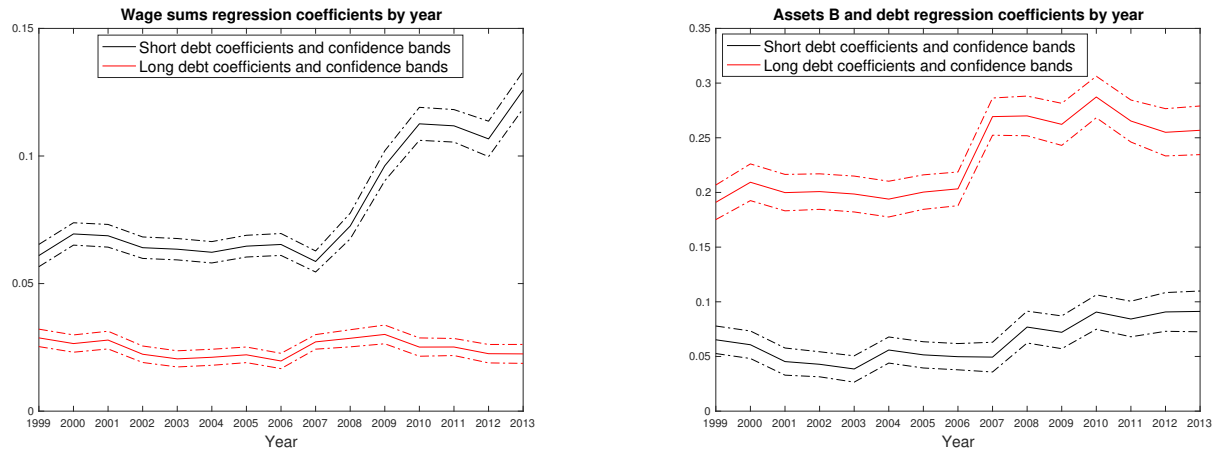
$$g^y_{it} = \alpha^y_t + \beta^y_{S,t} g^s_{it} + \beta^y_{L,t} g^b_{it} + X'_{it}\Gamma^y_t + u^y_{it}, \tag{2}$$

where X_{it} collects additional firm-level controls. We estimate the relationship year by year, so that $\beta^y_{S,t}$ and $\beta^y_{L,t}$ trace the strength of the conditional association between each type of borrowing and factor adjustment over time.

Equation (2) is intentionally symmetric in the two components of debt. Hence, the comparison between $\beta^y_{S,t}$ and $\beta^y_{L,t}$ tells us whether a given real margin is more closely associated with changes in short- or long-term borrowing, conditional on the other component of debt and the additional controls. This differs from an empirical specification based only on total leverage, which does not distinguish which part of the firm's liability structure moves together with employment or capital.

2.3 Debt Portfolios and Factor Adjustment

Figure 1 presents the central empirical result. Panel a reports the estimated coefficients from equation (2) when employment growth is the dependent variable. Panel b reports the corresponding estimates for capital growth.



(a) Employment

(b) Capital

Figure 1: Debt composition, employment, and capital. Panel (a) reports the year-specific coefficients on symmetric growth in short- and long-term debt from equation (2) when symmetric employment growth is the dependent variable. Panel (b) reports the corresponding coefficients when symmetric capital growth is the dependent variable. The bands report confidence intervals around the estimated coefficients.

The employment results reveal a stable and economically meaningful ordering. Throughout the sample, employment growth is more strongly associated with short-term debt growth than with long-term debt growth. The difference is already apparent before the financial crisis and becomes substantially larger around the Great Recession. In particular, the estimated association between employment and short-term borrowing rises sharply during this period, whereas the corresponding association with long-term borrowing changes much less.

The pattern is reversed for capital. As shown in Panel b, changes in capital are much more strongly associated

with changes in long-term debt than with changes in short-term debt. The long-debt coefficient exceeds the short-debt coefficient throughout the sample, and the difference is particularly pronounced from the onset of the financial crisis onward.

These results imply that aggregating the firm's liabilities into a single measure of debt masks an important dimension of firm behavior. Short- and long-term debt are not merely alternative labels for the same financing margin: in the data, they exhibit systematically different relationships with the two main factors of production. This distinction is especially relevant during periods of financial stress, when access to external finance is more likely to constrain real activity. The stronger connection between firm financing and employment during the Great Recession is consistent with the broader evidence on the employment effects of disruptions to credit supply and highly leveraged balance sheets [Chodorow-Reich, 2014, Giroud and Mueller, 2017].

The same ordering can be seen in simpler unconditional moments. Table 1 reports correlations in levels and in symmetric growth rates. In levels, log short-term debt has a correlation of 0.61 with log employment, compared with 0.47 for long-term debt. The ranking reverses for capital: the correlation of log capital with long-term debt is 0.65, compared with 0.50 for short-term debt. The distinction is even sharper in changes. The correlation between symmetric growth in short-term debt and employment is 0.11, more than twice the corresponding correlation with long-term debt, 0.05. For capital, the correlation with long-term debt is 0.22, compared with 0.09 for short-term debt.

Table 1: Debt Composition and Factor Inputs in the Data

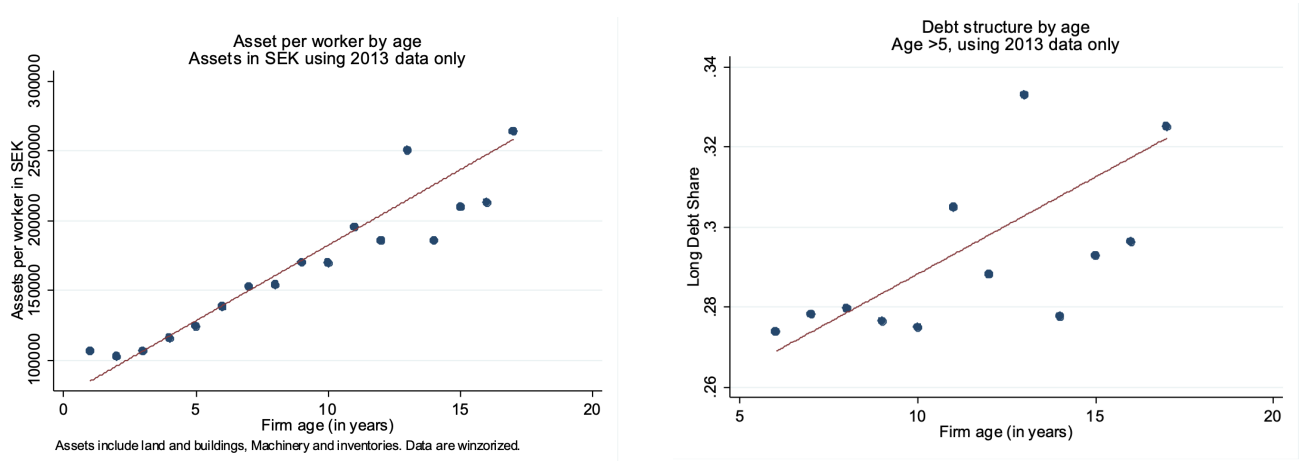
	Log levels	Symmetric growth rates
Short-term debt and employment	0.61	0.11
Long-term debt and employment	0.47	0.05
Short-term debt and capital	0.50	0.09
Long-term debt and capital	0.65	0.22

Notes: The table reports pairwise correlations in the Swedish firm-level data. The first column reports correlations between variables expressed in logs. The second column reports correlations between symmetric growth rates as defined in equation (1).

The evidence so far concerns the contemporaneous association between firms' financing and factor adjustment. We next show that debt composition is also systematically related to firms' evolution over the life cycle and to the capital intensity of their production structure.

2.4 Debt Composition over the Firm Life Cycle

We next examine how firms' production structure and debt composition evolve over the firm life cycle. Figure 2 plots capital per worker and the long-term debt share against firm age. Capital intensity rises markedly with age: older firms operate with substantially more capital per worker than younger firms. The composition of debt displays a similar, although more dispersed, age profile. Older firms tend to finance a larger share of their debt with long-term borrowing.



(a) Capital per worker

(b) Long-term debt share

Figure 2: Capital intensity and debt composition over the firm life cycle

Notes: Panel (a) plots average assets per worker against firm age. Panel (b) plots the average share of long-term debt in total debt against firm age for firms older than five years. Both panels use the 2013 cross section. The points represent age-group averages and the solid lines show the corresponding linear fits.

Taken together, the two panels point to a systematic evolution in firms' real and financial structure as they mature. Firms become increasingly capital intensive while simultaneously shifting their debt portfolios toward longer-term finance. The age profile of debt composition is qualitatively consistent with evidence that larger and more mature firms tend to borrow at longer maturities [Jungherr and Schott, 2021]. For our purposes, the important feature is that the two patterns occur together: the increase in the long-term debt share over the firm life cycle accompanies a pronounced increase in capital intensity.

2.5 Capital Intensity and Debt Composition

The life-cycle evidence suggests that the shift toward long-term borrowing is closely related to changes in firms' production structure. We examine this relationship more directly in Figure 3, which relates the long-term debt share to capital per worker after removing firm and time effects. The figure reveals a strong positive relationship: within firms, periods of greater capital intensity are systematically associated with a larger share of long-term debt.

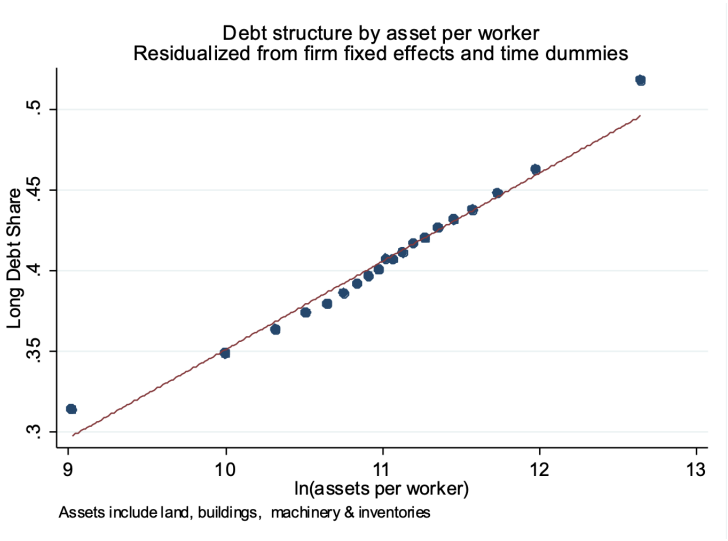


Figure 3: Capital intensity and debt composition

Notes: The figure plots the relationship between the long-term debt share and log assets per worker after removing firm fixed effects and time effects. The points represent grouped averages of the residualized variables and the solid line shows the corresponding linear fit.

This relationship provides a distinct piece of evidence from the age profiles in Figure 2. The increase in long-term borrowing among more capital-intensive firms is not driven solely by permanent differences across firms or by aggregate movements common to all firms. Instead, changes in firms' capital intensity are accompanied by changes in the composition of their debt portfolios. Together with the results in Section 2.3, this evidence points to a broader connection between firms' production structure and the maturity composition of their liabilities: employment-intensive adjustment is more closely associated with short-term borrowing, while greater capital intensity is associated with a larger role for long-term debt.

2.6 Robustness and Alternative Specifications

We assess the robustness of the debt–factor asymmetry along several dimensions. First, we re-estimate the baseline relationships across subsamples defined by firm size, age, sector, capital intensity, and the financial-crisis period. The ordering of the coefficients is preserved across these cuts of the data: employment remains more strongly associated with short-term debt, while capital remains more strongly associated with long-term debt.

Second, we progressively enrich the specification with additional balance-sheet variables and alternative sets of fixed effects. In particular, we control for transactions, cash, and equity, and consider specifications with year–sector effects, age and size controls, and firm fixed effects. These changes affect the magnitudes of the estimated coefficients but leave the relative association of the two debt instruments with employment and capital largely unchanged. The employment result is also consistent with evidence that liquid resources are particularly important for firms' employment decisions [Bacchetta et al., 2019].

Finally, we examine alternative measures of firms' real activity, including value added. These exercises lead to the same broad conclusion that the composition of borrowing contains information about firms' real adjustment that is obscured by measures of total debt alone. The full set of robustness results is reported in Appendix A.

2.7 Empirical Takeaways

Three empirical patterns guide the quantitative analysis. First, the maturity composition of debt is systematically related to firms' factor adjustment. Employment changes are more closely associated with short-term debt, whereas capital changes are more closely associated with long-term debt. This ordering is present throughout the sample and becomes especially pronounced around the Great Recession. Second, firms' financing structures evolve systematically over the life cycle: as firms age, they become more capital intensive and increase the share of long-term debt in their liabilities. Third, the relationship between capital intensity and debt composition extends beyond the age profile. Within firms, increases in capital per worker are associated with a larger long-term debt share after removing common time variation and persistent firm heterogeneity. The debt–factor asymmetry is also robust across alternative samples, controls, and fixed-effect specifications, as documented in Appendix A.

These facts point to economically distinct roles for short- and long-term finance in firms' real adjustment. They do not, however, identify the mechanism generating the observed relationships, since production and financing choices are jointly determined. The model developed next therefore asks whether differences in the maturity horizons, collateral requirements, adjustment costs, and default exposure of the two debt instruments can jointly account for the employment–short-debt and capital–long-debt asymmetries, as well as the evolution of debt composition as firms grow and mature.

3 Model

We develop a dynamic model of heterogeneous firms in which production and financing decisions are jointly determined. Firms differ in productivity, capital, and outstanding debt, and choose employment, investment, and the composition of external finance. The central feature of the environment is that firms finance their operations using two distinct debt instruments. Short-term debt provides within-period financing, allowing firms to meet expenditures before current revenues are realized, as in Jermann and Quadrini [2012], but exposes the firm to endogenous default risk. Long-term debt is collateralized by capital, in the spirit of Kiyotaki and Moore [1997], and provides a safer source of funding at the risk-free rate, subject to a collateral constraint. The composition of debt is therefore endogenous and interacts with the firm's choices of employment and capital accumulation.

The two debt instruments differ along several margins, giving rise to a dynamic portfolio choice. Greater reliance on short-term debt preserves collateral for future borrowing, but increases the firm's current repayment obligation, default probability, and short-term financing premium. Long-term borrowing, by contrast, reduces the need for short-term finance and hence current default exposure, but ties up collateral and raises the future cost of unsecured borrowing by reducing the assets available to short-term creditors in default. These forces also shape the evolution of the debt portfolio over the firm life cycle. Growing firms can use an expanding capital stock to support long-term borrowing and finance capital accumulation without proportionately increasing their exposure to short-term default risk. As long-term debt accumulates, however, further borrowing uses up collateral capacity and reduces the assets available to short-term creditors, worsening future short-term borrowing terms. Additional long-term finance therefore becomes progressively less attractive at the margin. Firms consequently rely more on short-term debt to meet additional financing needs, even though older and more capital-intensive firms hold larger stocks, and a larger portfolio share, of long-term debt.

3.1 Environment and Timing

The economy is populated by a continuum of production firms. Firms are ex ante identical but become heterogeneous in productivity, capital, and outstanding long-term debt. A firm enters period t with capital k_t and long-term debt b_t. At the beginning of the period, it observes its idiosyncratic productivity z_t. The firm's state is therefore

$$x_t \equiv (z_t, k_t, b_t). \tag{3}$$

Conditional on this state, the firm chooses employment n_t, next-period capital k_{t+1}, and next-period long-term debt b_{t+1}. These choices, together with the repayment of previously issued long-term debt and the firm's operating costs, determine its financing requirement within the period. Whenever internal resources and new long-term borrowing are insufficient to cover these expenditures, the firm raises short-term debt s_t. A firm borrowing s_t promises a repayment $p_t s_t$ at the end of the period, where p_t is determined competitively by short-term creditors and reflects the firm's repayment risk.

Production takes place using current capital and labor. After production decisions have been made, the firm draws an idiosyncratic revenue shock ϕ_t, which determines the fraction of planned output that translates into realized revenue. The shock can be interpreted as reflecting either a production shortfall or weak demand for the firm's output. The timing is important: ϕ_t is not known when employment, investment, and financing decisions are chosen. Consequently, the firm may find itself unable to meet its short-term obligation after revenues are realized.

If realized revenues are insufficient to repay short-term debt, the firm can liquidate part of the capital chosen for the following period. Capital liquidation is costly and is limited by the collateral requirement associated with long-term debt. If the firm cannot meet its short-term obligation even after exhausting the available liquidation margin, it defaults and exits.

The sequence of events within a period is therefore:

1. The firm enters period t with capital k_t and outstanding long-term debt b_t, and observes productivity z_t.
2. The firm jointly chooses employment n_t, next-period capital k_{t+1}, next-period long-term debt b_{t+1}, and short-term debt s_t. These choices determine expected dividends d_t.
3. Production takes place using (z_t, n_t, k_t).
4. The idiosyncratic revenue shock ϕ_t is realized.
5. The firm uses realized revenues to repay its short-term obligation. If repayment remains infeasible after exhausting the available liquidation margin, the firm defaults at the end of period t. The defaulting firm is resolved and exits at the beginning of period $t+1$.
6. Surviving firms enter period $t+1$ with their resulting capital stock and the long-term debt position b_{t+1}.

This timing generates the distinction between the two financing margins that is central to the analysis. Firms can use either source of borrowing to finance their expenditures, and the composition of short- and long-term debt is chosen jointly with employment and capital. Short-term borrowing is repaid within the period and exposes the firm to repayment and default risk once revenues are realized. Long-term borrowing is collateralized by capital, is carried forward on the firm's balance sheet, and affects both current borrowing capacity and the resources

available to creditors in future periods. We next describe the firm's production technology and the two debt instruments in detail.

3.2 The Firm

3.2.1 Production and Capital Adjustment

A firm enters period t with capital k_t and outstanding long-term debt b_t, and observes its idiosyncratic productivity z_t. Conditional on these state variables, the firm chooses employment n_t, next-period capital k_{t+1}, next-period long-term debt b_{t+1}, and short-term borrowing s_t. These choices determine expected dividends d_t. The financing constraints that jointly determine these choices are introduced below.

The firm produces output using a Cobb–Douglas technology with decreasing returns to scale,

$$y_t = z_t f(n_t, k_t) = z_t n_t^{\alpha} k_t^{\gamma}, \qquad \alpha > 0, \quad \gamma > 0, \quad \alpha + \gamma < 1, \tag{4}$$

where α and γ denote the output elasticities with respect to labor and capital, respectively. Current period capital is predetermined, whereas employment is chosen after the realization of z_t.

Log firm-level productivity follows a Gaussian AR(1) process,

$$\log z_{t+1} = \rho \log z_t + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_{\varepsilon}^2), \tag{5}$$

where $\rho \in (0,1)$ governs the persistence of firm-level productivity and ε_{t+1} is an idiosyncratic innovation with variance σ_{ε}^2. Hence, conditional on current productivity, $\log z_{t+1}$ is normally distributed.

Capital available for production in period $t+1$ is chosen in period t. Capital depreciates at rate δ, and reductions in the capital stock are partially irreversible. Let Δk_t denote the resources generated by the firm's capital adjustment:

$$\Delta k_t = \begin{cases} (1-\delta)k_t - k_{t+1}, & k_{t+1} \geq (1-\delta)k_t - \Omega k_t, \\ \Omega k_t + (1-\nu)\left[(1-\delta)k_t - \Omega k_t - k_{t+1}\right], & k_{t+1} < (1-\delta)k_t - \Omega k_t. \end{cases} \tag{6}$$

A fraction Ω of the firm's capital stock can therefore be liquidated without a loss. Beyond this liquid portion, capital can only be resold at a fraction $1-\nu$ of its value. The parameter $\nu \in [0,1]$ measures the degree of partial irreversibility: a larger ν implies a larger loss from liquidating installed capital. Such resale losses capture the limited redeployability of installed capital (see Khan and Thomas [2008]).

In addition to partial irreversibility, capital adjustment outside the frictionless region entails a quadratic cost, following Bloom [2009],

$$C(k_t, k_{t+1}) = \mathbf{1}_{\{k_{t+1} > (1-\delta)k_t \text{ or } k_{t+1} < (1-\delta)k_t - \Omega k_t\}} c \frac{(k_{t+1} - k_t)^2}{k_t}, \tag{7}$$

where $c > 0$ governs the magnitude of the quadratic adjustment cost.

The interval between $(1-\delta)k_t - \Omega k_t$ and $(1-\delta)k_t$ constitutes a frictionless liquidation margin. We interpret the amount Ωk_t as the liquid component of the firm's productive capital stock, playing a role similar to a cash buffer: the firm can draw down this portion of capital to meet expenditures without incurring either the resale loss associated with partial irreversibility or the convex adjustment cost. Reductions in capital beyond this liquid

buffer trigger both the resale loss governed by ν and the quadratic adjustment cost, with the latter applied to the full capital adjustment. Conversely, increases in capital above the depreciated capital stock, $(1-\delta)k_t$, are subject to the quadratic adjustment cost.

3.2.2 Long-Term Debt

Firms can use long-term debt to finance their expenditures. In the spirit of Kiyotaki and Moore [1997], long-term borrowing is secured by productive capital. We consider a small open economy, however, so the risk-free interest rate r is exogenous. Let b_t denote the stock of long-term debt outstanding at the beginning of period t and b_{t+1} the stock chosen for the following period. Long-term debt is fully collateralized and carries the risk-free interest rate r. Borrowing is subject to

$$b_{t+1} \leq \lambda k_{t+1}, \tag{8}$$

where $\lambda \in (0,1)$ denotes the fraction of next-period capital that can be pledged to long-term creditors.

Adjusting the stock of long-term debt is costly. In particular,

$$\psi(b_t, b_{t+1}) = \tau \left| b_{t+1} - b_t \right|, \tag{9}$$

where $\tau \geq 0$ governs the adjustment cost. This reduced-form cost captures issuance and restructuring costs that make debt positions costly to adjust, consistent with the evidence in Altınkılıç and Hansen [2000] and with the issuance-cost mechanism in Jungherr and Schott [2021].

In addition, maintaining a positive long-term debt position entails the fixed cost

$$c_{ld}\mathbf{1}\{b_{t+1} > 0\}. \tag{10}$$

We interpret c_{ld} as a reduced-form fixed participation cost associated with accessing and maintaining long-term finance. Fixed financing costs are a standard ingredient in dynamic capital-structure models and help account for the infrequent adjustment of firms' debt positions [Fischer et al., 1989, Leary and Roberts, 2005].

Thus, net resources generated by long-term borrowing in period t are $b_{t+1} - (1+r)b_t$, net of the adjustment and fixed costs in equations (9) and (10). Because borrowing capacity depends on k_{t+1}, the firm's investment and long-term financing decisions are jointly determined.

3.2.3 Short-Term Debt

Firms must finance their production and investment decisions before current revenues are realized. As in Jermann and Quadrini [2012][1], this timing creates a working-capital requirement within the period. In our environment, firms jointly choose short- and long-term debt as part of their financing portfolio, with the two instruments differing in maturity, collateral requirements, adjustment costs, and exposure to default risk.

Let s_t denote short-term borrowing in period t. For each unit borrowed, the firm promises to repay p_t at the end of the period, so that its total short-term obligation is $p_t s_t$. The price p_t is firm specific and reflects the risk that the firm may be unable to repay after revenues are realized. Its determination is described in Section 3.4.

[1] Working-capital financing has a long tradition in quantitative macroeconomics; see, among others, Fuerst [1992], Neumeyer and Perri [2005], Christiano et al. [2005], Ravenna and Walsh [2006], and Mendoza [2010]. Neumeyer and Perri [2005] study this mechanism in a small open economy, while Mendoza [2010] combines working-capital financing with collateral constraints.

Given the firm's choices of employment, capital, and long-term debt, the short-term financing requirement satisfies

$$s_t = (1+r)b_t + w_t n_t + C(k_t, k_{t+1}) + c_f + c_{ld}\mathbf{1}\{b_{t+1} > 0\} + \psi(b_t, b_{t+1}) - b_{t+1} - \Delta k_t, \tag{11}$$

where c_f denotes the fixed operating cost. When the right-hand side of equation (11) is negative, the firm does not require short-term borrowing. In this case, $p_t = 1$, and the excess resources, $-s_t$, are distributed to shareholders as dividends.

Equation (11) links the firm's expenditures to the resources available from short- and long-term borrowing and from capital adjustment. The two debt instruments are therefore chosen jointly as part of the firm's financing portfolio.

For given employment and capital choices, additional long-term borrowing allows the firm to reduce short-term borrowing, net of the costs associated with adjusting and maintaining the long-term debt position. At the same time, as shown below, a larger long-term debt position reduces the recovery value available to short-term creditors and therefore affects the price of short-term borrowing. The composition of debt consequently reflects the trade-off between these financing margins.

3.3 Revenue Risk, Financial Distress, and Exit

After production and financing decisions have been made, the firm draws an idiosyncratic revenue shock ϕ_t with density $g(\phi)$ and distribution function $G(\phi)$ on $[\phi_l, \phi_m]$. The revenue shock ϕ_t is independent of the productivity process. The shock captures ex post uncertainty in the fraction of planned output that translates into realized revenue: it may reflect either a production shortfall or weak demand for the firm's output.[2] Realized revenue is therefore

$$\phi_t y_t = \phi_t z_t n_t^{\alpha} k_t^{\gamma}. \tag{12}$$

At the end of the period, the firm must repay its short-term obligation $p_t s_t$. If realized revenue is insufficient, the firm can liquidate part of the capital stock k_{t+1} chosen for the following period to meet the remaining obligation. The liquid component Ωk_{t+1} can be drawn down without incurring either the resale loss associated with partial irreversibility or the convex capital-adjustment cost. Liquidation beyond this buffer is subject to both the resale loss governed by ν and the convex adjustment cost, and must leave sufficient capital to satisfy the collateral constraint on long-term debt.

Let $f^s(k_{t+1}, b_{t+1})$ denote the maximum net proceeds that can be generated through capital liquidation while preserving sufficient capital to collateralize long-term debt. For any reduction from k_{t+1} to $\tilde{k}_{t+1} \leq k_{t+1}$, let $\mathcal{L}(k_{t+1}, \tilde{k}_{t+1})$ denote the net resources generated by the liquidation:

$$\mathcal{L}(k_{t+1}, \tilde{k}_{t+1}) = \begin{cases} k_{t+1} - \tilde{k}_{t+1}, & \tilde{k}_{t+1} \geq k_{t+1} - \Omega k_{t+1}, \\ \Omega k_{t+1} + (1-\nu)\left[k_{t+1} - \Omega k_{t+1} - \tilde{k}_{t+1}\right] - c\,\dfrac{\left(\tilde{k}_{t+1} - k_{t+1}\right)^2}{k_{t+1}}, & \tilde{k}_{t+1} < k_{t+1} - \Omega k_{t+1}. \end{cases}$$

[2] Related models let firms choose inputs before the realization of idiosyncratic shocks that determine revenues and default risk. Cooke and Damjanovic [2020], for example, introduce an idiosyncratic demand shock after labor has been hired and production has taken place, while Arellano et al. [2019] study firm-level uncertainty when production decisions precede the realization of idiosyncratic shocks.

$$f^s(k_{t+1}, b_{t+1}) = \max_{\widetilde{k}_{t+1} \geq b_{t+1}/\lambda} \mathcal{L}(k_{t+1}, \widetilde{k}_{t+1}), \tag{13}$$

The constraint $\tilde{k}_{t+1} \geq b_{t+1}/\lambda$ ensures that sufficient capital remains to collateralize long-term debt.

The realization of ϕ_t gives rise to three outcomes. If

$$p_t s_t \leq \phi_t y_t, \tag{14}$$

realized revenue is sufficient to meet the short-term obligation, and the firm repays without liquidating capital. If instead

$$\phi_t y_t < p_t s_t \leq \phi_t y_t + f^s(k_{t+1}, b_{t+1}), \tag{15}$$

realized revenue alone is insufficient, but the firm remains solvent by liquidating part of its capital stock. It then enters the next period with the reduced capital stock $\tilde{k}_{t+1}$. For $\phi_t^e \leq \phi < \phi_t^r$, let $\tilde{k}_{t+1}(\phi)$ denote the remaining capital stock after the firm liquidates just enough capital to meet its short-term obligation. It is implicitly defined by

$$\mathcal{L}\Big(k_{t+1}, \tilde{k}_{t+1}(\phi)\Big) = p_t s_t - \phi y_t.$$

Finally, if

$$p_t s_t > \phi_t y_t + f^s(k_{t+1}, b_{t+1}), \tag{16}$$

even the maximum feasible liquidation proceeds are insufficient to meet the short-term obligation. The firm therefore defaults on its short-term debt and exits.

3.4 Short-Term Creditors and Endogenous Debt Prices

Short-term credit is supplied competitively by risk-neutral lenders. Over the intraperiod horizon of a short-term loan, lenders are indifferent between providing short-term credit and holding the funds in the risk-free asset, including long-term debt, since no interest accrues within the period. Competition therefore requires the expected payoff from a short-term loan to equal the amount advanced, s_t.[3]

In the event of default, long-term creditors have priority over short-term creditors. Long-term debt is secured by pledged capital and is repaid first, which makes it effectively risk-free and allows firms to borrow at the exogenous risk-free rate r. Short-term creditors are residual claimants on the liquidation value that remains after the collateral requirement associated with long-term debt has been satisfied. As defined in equation (13), $f^s(k_{t+1}, b_{t+1})$ is the maximum net amount that can be generated through capital liquidation while preserving sufficient collateral for long-term debt. In default, short-term creditors recover only a fraction ζ of this liquidation value and can recover at most the principal of the loan. Their recovery value is therefore

$$\mathcal{R}_t = \min\left\{s_t,\, \zeta f^s(k_{t+1}, b_{t+1})\right\}. \tag{17}$$

From equation (16), the revenue-shock realization at which the firm is just able to avoid default is[4]

[3] Competitive pricing of defaultable corporate debt based on expected repayment and recovery is standard in quantitative models of firm financing; see, for example, Arellano et al. [2019] and Jungherr and Schott [2021].

[4] Throughout, thresholds are interpreted relative to the support $[\phi_l, \phi_m]$ of the idiosyncratic revenue shock. If a threshold lies outside this support, the corresponding default, liquidation, or repayment region is empty or contains the entire support, as appropriate.

$$\phi_t^e = \frac{p_t s_t - f^s(k_{t+1}, b_{t+1})}{y_t}. \tag{18}$$

For realizations $\phi_t < \phi_t^e$, the firm defaults and short-term creditors receive $\mathcal{R}_t$. For $\phi_t \geq \phi_t^e$, the firm repays the promised amount $p_t s_t$.[5] The lenders' zero-profit condition is therefore

$$s_t = \int_{\phi_l}^{\phi_t^e} \min\left\{s_t,\, \zeta f^s(k_{t+1}, b_{t+1})\right\} g(\phi)\, d\phi + \int_{\phi_t^e}^{\phi_m} p_t s_t\, g(\phi)\, d\phi, \tag{19}$$

or, equivalently,

$$s_t = \min\left\{s_t,\, \zeta f^s(k_{t+1}, b_{t+1})\right\} G\left(\phi_t^e\right) + p_t s_t \left[1 - G\left(\phi_t^e\right)\right]. \tag{20}$$

Since the default threshold in equation (18) itself depends on p_t, equation (20) implicitly determines the firm-specific short-term debt price. We denote this pricing schedule by

$$p_t = H\left(s_t,\, y_t,\, k_{t+1},\, b_{t+1}\right), \tag{21}$$

where $y_t = z_t f(n_t, k_t)$. The price of short-term credit is therefore firm specific and reflects both the probability of default and the recovery value available to short-term creditors. Because both objects depend on the firm's state and policy choices, p_t varies across firms with different productivity, capital, and long-term debt positions. Greater short-term borrowing raises the promised repayment and, other things equal, increases default risk. A larger capital position increases the resources that can be generated through liquidation, strengthening creditor recovery. Greater long-term borrowing works in the opposite direction by reducing the capital that can be liquidated while satisfying the collateral requirement. Hence, the firm's debt portfolio affects not only the quantity of short-term borrowing but also its price. Consequently, even in a stationary equilibrium, heterogeneity in firms' states generates a non-degenerate cross-sectional distribution of short-term debt prices.

A simple two-state version of the revenue shock makes these pricing forces particularly transparent. In that case, the short-term debt price can be written explicitly as a function of the probability of repayment, the amount borrowed, and creditor recovery. The resulting comparative statics show directly that the price of short-term debt rises with borrowing and falls with repayment probability and liquidation value. We derive these results formally in Appendix **??**.

3.5 Firm Value and Recursive Problem

At the beginning of period t, after observing productivity z_t, the firm's state is (z_t, k_t, b_t). The firm chooses employment n_t, next-period capital k_{t+1}, next-period long-term debt b_{t+1}, and short-term borrowing s_t, and thereby determines expected dividends d_t. These choices are made before the realization of the revenue shock ϕ_t. The firm chooses its policies to maximize the expected discounted value of dividends to shareholders.

It is useful to distinguish two revenue thresholds. The default threshold ϕ_t^e, defined in equation (18), is the lowest revenue realization for which the firm can meet its short-term obligation after exhausting the available capital-liquidation margin. Define also

$$\phi_t^r = \frac{p_t s_t}{y_t}, \tag{22}$$

[5] The short-term debt pricing condition is relevant only when $s_t > 0$. When $s_t = 0$, there is no short-term debt to price, so the value of p_t is immaterial.

as the revenue realization above which the firm can repay its short-term obligation without liquidating capital. Hence, $\phi_t^e \leq \phi_t^r$. For realizations $\phi \geq \phi_t^r$, realized revenue is sufficient to repay the short-term obligation without liquidating capital. Let $\widetilde{d}_t(\phi)$ denote the realized dividend. It is given by

$$\widetilde{d}_t(\phi) = \phi y_t - p_t s_t \geq 0.$$

For $\phi_t^e \leq \phi < \phi_t^r$, the firm remains solvent only by liquidating part of its chosen capital stock. All available resources are used to meet the short-term obligation, so the realized dividend is zero. Thus, conditional on survival,

$$\widetilde{d}_t(\phi) = \begin{cases} \phi y_t - p_t s_t, & \phi \geq \phi_t^r, \\ 0, & \phi_t^e \leq \phi < \phi_t^r. \end{cases}$$

The expected dividend determined before the realization of the revenue shock is therefore

$$d_t = \int_{\phi_t^r}^{\phi_m} (\phi y_t - p_t s_t)\, g(\phi)\, d\phi.$$

The revenue realization also determines the capital stock carried into the following period. For $\phi \geq \phi_t^r$, the firm enters period $t+1$ with its chosen capital stock k_{t+1}. For $\phi_t^e \leq \phi < \phi_t^r$, the firm remains solvent but liquidates part of its capital and enters the next period with $\widetilde{k}_{t+1}(\phi) \leq k_{t+1}$. For $\phi < \phi_t^e$, the firm defaults and exits.

Let $V^e(k_{t+1}, b_{t+1}, s_t)$ denote shareholder value upon default. After long-term debt has been repaid, short-term creditors receive the recovery value defined in equation (17). The remaining liquidation value accrues to shareholders. Let $V^e(k_{t+1}, b_{t+1}, s_t)$ denote the value accruing to shareholders upon resolution of a defaulting firm at the beginning of period $t+1$.

$$V^e(k_{t+1}, b_{t+1}, s_t) = f^s(k_{t+1}, b_{t+1}) - \mathcal{R}_t = f^s(k_{t+1}, b_{t+1}) - \min\left\{s_t,\, \zeta f^s(k_{t+1}, b_{t+1})\right\}. \tag{23}$$

The firm's recursive problem is

$$\begin{aligned} V(z_t, k_t, b_t) = \max_{n_t, k_{t+1}, b_{t+1}, s_t} \Bigg\{ d_t + \beta \Bigg[& \int_{\phi_t^r}^{\phi_m} \mathbb{E}_t V\left(z_{t+1}, k_{t+1}, b_{t+1}\right) g(\phi)\, d\phi \\ & + \int_{\phi_t^e}^{\phi_t^r} \mathbb{E}_t V\left(z_{t+1}, \widetilde{k}_{t+1}(\phi), b_{t+1}\right) g(\phi)\, d\phi \\ & + \int_{\phi_l}^{\phi_t^e} V^e\left(k_{t+1}, b_{t+1}, s_t\right) g(\phi)\, d\phi \Bigg] \Bigg\}, \end{aligned} \tag{24}$$

where $\mathbb{E}_t$ integrates over next-period productivity z_{t+1} conditional on z_t. The maximization is subject to the short-term financing constraint in equation (11), the collateral constraint in equation (8), the capital-adjustment technology, and the endogenous short-term debt pricing schedule in equation (21).

Entrants. New firms enter before the realization of their initial productivity. An entrant begins without inherited long-term debt and draws its initial capital stock from

$$k^e \sim U[k_{\min}, k_{\max}]. \tag{25}$$

Let $\Pi^e(z)$ denote the ergodic distribution associated with the productivity process in equation (5), from which entrants draw their initial productivity. Conditional on the realizations of k^e and z, an entrant faces the same production, financing, and default problem as an incumbent firm, with initial state $(z, k^e, 0)$.

Entry is not governed by a free-entry condition. Instead, firms that exit through default are replaced by new entrants, as described in Section 4.6.

3.6 Stationary Equilibrium

We close the model by specifying the labor market and the evolution of the cross-sectional distribution of firms. The economy is populated by a unit measure of workers who supply labor inelastically. Hence, aggregate labor supply is equal to one. The wage w_t is determined competitively and adjusts so that aggregate labor demand equals this fixed supply. The final good is the numeraire. Consistent with the small-open-economy assumption, the risk-free interest rate r is exogenous and long-term credit is supplied elastically at this rate.

Let

$$x = (z, k, b) \in \mathcal{X} \equiv \mathcal{Z} \times \mathcal{K} \times \mathcal{B}$$

denote a firm's beginning-of-period state, and let μ_t denote the measure of active firms over $\mathcal{X}$. For a given wage, denote the firm's policy vector by

$$\sigma(x) = \left\{ n(x),\, k^+(x),\, b^+(x),\, s(x) \right\},$$

where k^+ and b^+ denote next-period capital and long-term debt, respectively. Expected dividends are implied by these policies and the distribution of the revenue shock as described in Section 3.5.

Given these policies, let $\mathcal{Q}(A \mid x; w)$ denote the probability that a firm currently in state x survives and enters the following period in a state belonging to the Borel set $A \subseteq \mathcal{X}$. In particular,

$$\mathcal{Q}(A \mid x; w) = \int_{\mathcal{Z}} \left[\int_{\phi^r(x)}^{\phi_m} \mathbf{1}\left\{ \left(z^+, k^+(x), b^+(x)\right) \in A \right\} g(\phi)\, d\phi + \int_{\phi^e(x)}^{\phi^r(x)} \mathbf{1}\left\{ \left(z^+, \widetilde{k}^+(x,\phi), b^+(x)\right) \in A \right\} g(\phi)\, d\phi \right] \Pi(dz^+ \mid z), \tag{26}$$

where $\Pi(\cdot \mid z)$ is the transition distribution of productivity. The missing probability mass, $1 - \mathcal{Q}(\mathcal{X} \mid x; w)$, is the firm's probability of default and exit.

Exiting firms are replaced by new entrants. Let η^e denote the distribution of entrant states. Entrants begin without inherited long-term debt, their initial capital is drawn uniformly from $[k_{\min}, k_{\max}]$, and their initial productivity is drawn from the stationary distribution Π^e associated with the productivity process. Hence, for any $A \subseteq \mathcal{X}$,

$$\eta^e(A) = \frac{1}{k_{\max} - k_{\min}} \int_{k_{\min}}^{k_{\max}} \int_{\mathcal{Z}} \mathbf{1}\{(z, k, 0) \in A\} \Pi^e(dz)\, dk. \tag{27}$$

Let M_t^e denote the mass of entrants in period t. Since entrants replace exiting firms, their mass is

$$M_t^e = \mu_t(\mathcal{X}) - \int_{\mathcal{X}} \mathcal{Q}(\mathcal{X} \mid x; w_t) \mu_t(dx). \tag{28}$$

The law of motion of the firm distribution is therefore

$$\mu_{t+1}(A) = \int_{\mathcal{X}} \mathcal{Q}(A \mid x; w_t)\, \mu_t(dx) + M_t^e \eta^e(A), \qquad A \subseteq \mathcal{X}. \tag{29}$$

A stationary distribution μ^* is invariant under equation (29). Although the cross-sectional distribution is constant in a stationary equilibrium, individual firms continue to experience productivity and revenue shocks, adjust employment, capital, and their debt portfolios, and enter or exit.

Definition. A stationary competitive equilibrium is a collection

$$\mathcal{E}^* = \{V^*, \sigma^*, H^*, w^*, \mu^*, M^{e,*}\} \tag{30}$$

such that:

1. *Firm optimization.* Given the equilibrium wage w^* and the exogenous risk-free rate r, the value function V^* and policy functions

   $$\sigma^*(x) = \left\{n^*(x), k^{+*}(x), b^{+*}(x), s^*(x)\right\}$$

   solve the firm's recursive problem in equation (24), subject to all financing, collateral, and capital-adjustment constraints. Expected dividends are determined by these optimal policies as described in Section 3.5.

2. *Short-term credit pricing.* For every firm state x, the firm-specific short-term debt price

   $$p^*(x) = H^*\left(s^*(x), y^*(x), k^{+*}(x), b^{+*}(x)\right)$$

   satisfies the competitive lenders' zero-profit condition in equation (20).

3. *Stationarity of the firm distribution.* For every Borel set $A \subseteq \mathcal{X}$,

   $$\mu^*(A) = \int_{\mathcal{X}} \mathcal{Q}(A \mid x; w^*)\, \mu^*(dx) + M^{e,*} \eta^e(A), \tag{31}$$

   where

   $$M^{e,*} = \mu^*(\mathcal{X}) - \int_{\mathcal{X}} \mathcal{Q}(\mathcal{X} \mid x; w^*) \mu^*(dx). \tag{32}$$

   Thus, in the stationary equilibrium, the mass of entering firms equals the mass of firms exiting through default.

4. *Labor-market clearing.* The equilibrium wage w^* adjusts so that aggregate labor demand exhausts the unit measure of workers:

   $$\int_{\mathcal{X}} n^*(x)\, \mu^*(dx) = 1. \tag{33}$$

Thus, the wage and the cross-sectional distribution of firms are jointly determined in equilibrium. The wage affects firms' employment, investment, financing, and default decisions, which in turn determine the stationary distribution μ^*; the resulting distribution of labor demand must, in turn, satisfy equation (33). The world risk-free rate remains exogenous throughout.

4 Debt Portfolio Choice and Real Decisions

This section characterizes the economic forces governing firms' debt portfolios and their interaction with employment and capital decisions. The central trade-off arises because short- and long-term debt differ in their maturity horizons, collateral requirements, and exposure to default risk. Greater reliance on short-term debt raises the firm's current repayment obligation, increasing the price of short-term credit and the probability of default and exit. At the same time, it reduces long-term borrowing and therefore leaves the firm with a smaller stock of inherited long-term debt in the following period, improving future short-term borrowing terms and reducing future default risk. Greater reliance on long-term debt has the opposite intertemporal consequences. It reduces the firm's current need for short-term finance and hence lowers its current short-term debt price and default risk, but leaves the firm with a larger stock of long-term debt in the following period, which worsens the recovery position of future short-term creditors and raises future financing costs and default risk. The firm's debt portfolio therefore balances current against future exposure to financial distress.

We first characterize the marginal conditions governing firms' financing and production decisions and then use the associated policy functions to trace these mechanisms across firm states. Additional derivations and the firm's optimality conditions are provided in Appendix B. Section 5 presents the calibration and quantitative implications of the model.

4.1 Financing Needs and Short-Term Credit Terms

The financing requirement allows us to express short-term borrowing as a function of the firm's state and its choices of employment, capital, and long-term debt. For a firm with state $x_t = (z_t, k_t, b_t)$, define

$$\mathcal{S}\left(x_t, n_t, k_{t+1}, b_{t+1}\right) \equiv (1+r)b_t + w_t n_t + C(k_t, k_{t+1}) + c_f + c_{ld}\mathbf{1}\{b_{t+1} > 0\} + \psi(b_t, b_{t+1}) - b_{t+1} - \Delta k_t. \tag{34}$$

Hence,

$$s_t = \mathcal{S}\left(x_t, n_t, k_{t+1}, b_{t+1}\right). \tag{35}$$

Short-term borrowing is therefore the residual financing requirement implied by the firm's production, investment, and long-term borrowing decisions. The repayment factor on this debt is itself endogenous. Substituting equation (35) into the lenders' pricing schedule gives

$$p_t = H\left(\mathcal{S}\left(x_t, n_t, k_{t+1}, b_{t+1}\right), y_t, k_{t+1}, b_{t+1}\right). \tag{36}$$

Thus, employment, capital, and long-term debt affect the firm's financing terms not only through the amount of short-term borrowing they require, but also through their effects on default risk and creditor recovery.

Away from the discrete long-debt participation margin and the kinks in the adjustment technologies, the effect of the firm's choices on its short-term financing requirement follows directly from equation (35). In particular,

$$\frac{\partial s_t}{\partial n_t} = w_t > 0, \tag{37}$$

$$\frac{\partial s_t}{\partial b_{t+1}} = -1 + \frac{\partial \psi(b_t, b_{t+1})}{\partial b_{t+1}}, \tag{38}$$

$$\frac{\partial s_t}{\partial k_{t+1}} = \frac{\partial C(k_t, k_{t+1})}{\partial k_{t+1}} - \frac{\partial \Delta k_t}{\partial k_{t+1}}. \tag{39}$$

Employment increases the firm's residual financing need directly through the wage bill. Long-term borrowing provides current resources and reduces the need for short-term finance, net of the cost of adjusting the long-debt position (for $\tau < 1$, away from the adjustment-cost kink). Capital affects the financing requirement through investment expenditures, adjustment costs, and the net resources generated by capital adjustment. These direct effects on short-term borrowing are only one part of the financing problem, however, because the amount of short-term debt and the assets available for creditor recovery jointly determine the terms of short-term credit and the probability of default.

To characterize this second margin, consider the interior region in which short-term creditors recover less than the principal of the loan in default, so that

$$\mathcal{R}_t = \zeta f^s(k_{t+1}, b_{t+1}) < s_t.$$

Using the default threshold in equation (18), define the lenders' zero-profit condition as

$$F(p_t, s_t, y_t, k_{t+1}, b_{t+1}) \equiv R_t G(\phi_t^e) + p_t s_t \left[1 - G(\phi_t^e)\right] - s_t = 0, \tag{40}$$

where

$$\phi_t^e = \frac{p_t s_t - f^s(k_{t+1}, b_{t+1})}{y_t}$$

Let $G_t \equiv G(\phi_t^e)$ and $g_t \equiv g(\phi_t^e)$. Using equation (40), the relevant partial derivatives, holding p_t fixed, are are given below. The corresponding derivations are provided in Appendix B.4.

$$F_s = -\frac{R_t}{s_t} G_t - \frac{p_t(p_t s_t - R_t)}{y_t} g_t < 0, \tag{41}$$

$$F_y = \frac{(p_t s_t - R_t)\left[p_t s_t - f^s(k_{t+1}, b_{t+1})\right]}{y_t^2} g_t > 0, \tag{42}$$

$$F_k = \left[\zeta G_t + \frac{p_t s_t - \mathcal{R}_t}{y_t} g_t\right] f_k^s > 0. \tag{43}$$

$$F_b = \left[\zeta G_t + \frac{p_t s_t - \mathcal{R}_t}{y_t} g_t\right] f_b^s < 0. \tag{44}$$

where $f_k^s \equiv \partial f^s / \partial k_{t+1} \geq 0$ and $f_b^s \equiv \partial f^s / \partial b_{t+1} \leq 0$. The signs reflect the fact that greater output improves repayment capacity, while greater capital increases and greater long-term debt reduces the liquidation value available to short-term creditors.

On a locally regular branch of the pricing equation for which $F_p > 0$, the implicit-function theorem therefore gives

$$H_s > 0, \qquad H_y < 0, \qquad H_k < 0, \qquad H_b > 0. \tag{45}$$

Thus, greater short-term borrowing requires a higher repayment factor, while greater output and capital improve the terms of short-term credit. Holding short-term borrowing fixed, a larger long-term debt position instead raises the repayment factor because it weakens short-term creditors' recovery position. In the full-recovery region, by contrast, short-term creditors face no expected loss and the repayment factor is $p_t = 1$.

The interaction between the two debt instruments can now be seen directly. Holding employment and next-

period capital fixed, the effect of long-term borrowing on the short-term repayment factor is

$$\left.\frac{\partial p_t}{\partial b_{t+1}}\right|_{n_t,k_{t+1}} = H_s \frac{\partial s_t}{\partial b_{t+1}} + H_b. \tag{46}$$

For $\tau < 1$, the first term is negative: additional long-term borrowing provides current resources and reduces the firm's need for short-term finance. The second term is positive: the same increase in long-term debt reduces the residual assets available to short-term creditors. Long-term borrowing therefore improves current funding through one channel while worsening the recovery position of short-term creditors through another. Appendix B.5 derives the total response of the short-term debt price to the firm's employment, capital, and long-term borrowing choices.

4.2 The Short- versus Long-Term Debt Trade-off

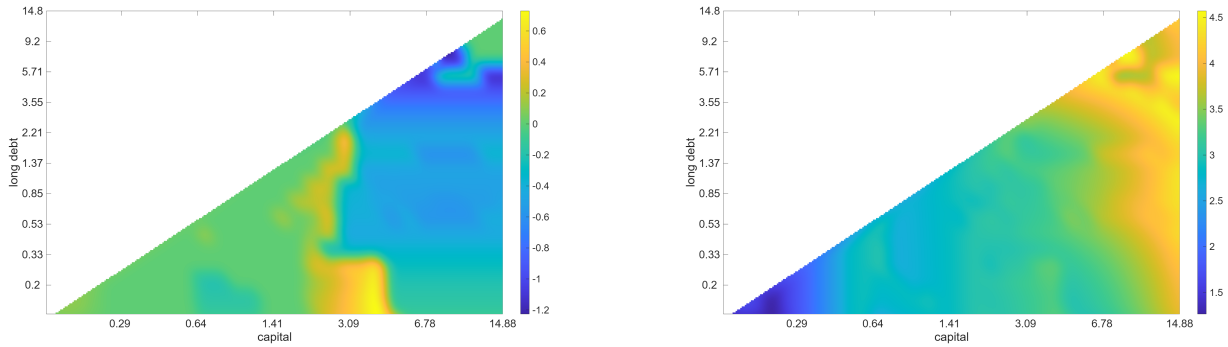
The choice between short- and long-term debt reflects a trade-off between their different implications for current funding, creditor recovery, and future financing conditions. Consider first an increase in long-term borrowing, b_{t+1}. For given employment and capital choices, additional long-term debt provides current resources and allows the firm to finance a smaller share of its expenditures with short-term credit. This reduces the firm's exposure to the repayment risk associated with short-term borrowing. At the same time, however, a larger long-term debt position encumbers a greater share of the firm's assets. Holding short-term borrowing fixed, this reduces the recovery value available to short-term creditors and therefore raises the repayment factor required on short-term credit. Thus, even within the current period, greater reliance on long-term debt provides a funding benefit while weakening the recovery position supporting short-term borrowing.

Long-term debt also carries a dynamic cost. The choice of b_{t+1} becomes the firm's inherited debt position in the following period. A larger inherited debt stock increases future debt-service obligations and, conditional on future capital, reduces the assets available to short-term creditors in default. Greater long-term borrowing today can therefore worsen the terms of short-term finance and increase financial distress in future periods. The dynamic cost of long-term debt is related to the debt-overhang mechanism in Jungherr and Schott [2021], but operates here through the effect of senior collateralized debt on the recovery value and pricing of future short-term credit.

Tilting the portfolio toward short-term debt entails the corresponding trade-off. When short-term finance is relatively attractive, the firm optimally chooses a smaller long-term debt position and makes greater use of short-term borrowing. Doing so preserves more of the firm's assets from long-term encumbrance and leaves less inherited debt to be carried into the following period, improving future financing conditions. The cost is greater current exposure to defaultable short-term credit, which increases the firm's repayment burden and its sensitivity to the realization of the revenue shock. The optimal debt portfolio therefore balances the current funding advantage of long-term borrowing against both its effect on current creditor recovery and the future cost of carrying long-term debt forward.

The portfolio trade-off is reflected directly in firms' financing policies. Figure 4 plots the optimal change in long-term debt and the optimal amount of short-term borrowing over the firm's state space, for a given level of productivity and evaluated at the benchmark calibration. Both policies vary substantially with the firm's capital stock and inherited long-term debt. The composition of financing is therefore state dependent: firms optimally adjust their use of short- and long-term debt as the relative current and future costs of the two instruments

change across balance-sheet positions.



(a) Change in long-term debt

(b) Short-term borrowing

Figure 4: Debt portfolio policies. Panel (a) shows the optimal change in long-term debt, $b_{t+1} - b_t$, and Panel (b) shows optimal short-term borrowing, s_t, as functions of the firm's current capital and inherited long-term debt, for a given level of productivity.

The two panels also make clear that the substitution between the debt instruments is neither mechanical nor one-for-one. A firm's preferred debt composition is chosen jointly with employment and capital, while adjustment costs, collateral capacity, default risk, and the endogenous terms of short-term credit all vary with the state. When short-term borrowing becomes relatively attractive, the firm can deliberately choose a smaller long-term debt position and rely more heavily on short-term finance; when the benefits of collateralized long-term borrowing dominate, it tilts the portfolio in the opposite direction. The policy functions therefore summarize an endogenous portfolio choice rather than a fixed assignment of particular expenditures to particular debt instruments.

4.3 Short-Term Credit Terms and Default Risk

The price of short-term credit is endogenous to the firm's balance sheet and production decisions. For an interior default threshold, the firm's probability of default is

$$\Pr_t(\text{default} \mid x_t) = G(\phi_t^e) = G\left(\frac{p_t s_t - f^s(k_{t+1}, b_{t+1})}{y_t}\right). \tag{47}$$

Because the default threshold itself depends on p_t, the lenders' zero-profit condition and the default threshold jointly determine the firm-specific short-term debt price,

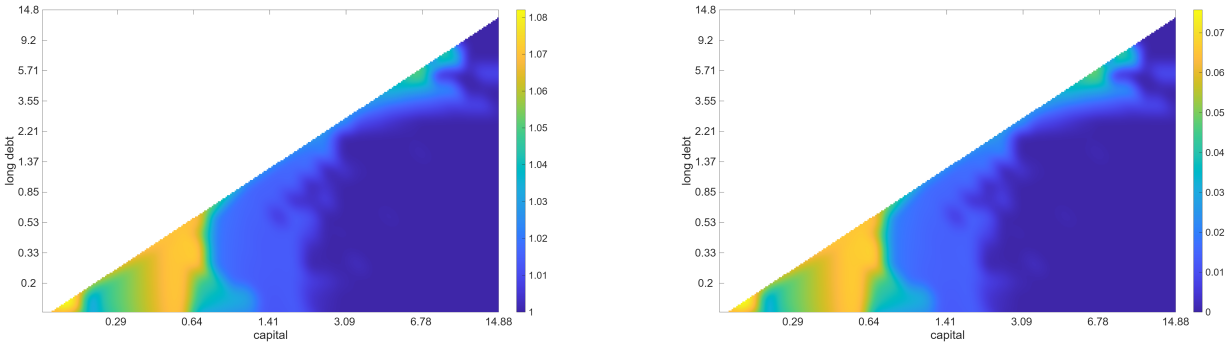
$$p_t = H(s_t, y_t, k_{t+1}, b_{t+1}), \tag{48}$$

and the associated probability of default. Competitive pricing of defaultable debt according to expected repayment and recovery follows the standard logic in Arellano et al. [2019].

The short-term debt price is closely related to default risk, but also reflects the recovery value available to lenders. Greater short-term borrowing raises the firm's repayment obligation, while greater output increases the resources available for repayment. Capital strengthens the recovery position of short-term creditors, whereas inherited long-term debt works in the opposite direction because long-term creditors have priority over short-term claims. Consequently, firms with otherwise similar financing needs can face different short-term debt prices

depending on the composition of their balance sheets.

Figure 5 illustrates this state dependence. It plots the short-term debt price and the corresponding probability of default over the firm's state space, for a given level of productivity and evaluated at the benchmark calibration. The two policy functions display closely related patterns, although the relationship is not mechanical because the debt price depends on recovery conditional on default as well as on the probability of default itself. In the policy functions, states with weaker recovery positions tend to be associated with both higher short-term debt prices and greater default risk.



(a) Short-term debt price

(b) Probability of default

Figure 5: Short-term credit terms and default risk. Panel (a) shows the firm-specific price of short-term debt, p_t, and Panel (b) shows the corresponding probability of default, as functions of the firm's current capital and inherited long-term debt, for a given level of productivity. Policy functions are evaluated at the benchmark calibration.

The heterogeneity in short-term credit terms is important for firms' real decisions. Employment, capital, and debt composition alter both the amount of external finance required and the price at which short-term funds can be obtained. Firms therefore face state-dependent marginal financing costs, which provide the link between the debt-portfolio mechanism and the employment and capital decisions considered next.

4.4 Financing, Employment, and Capital

The two production margins interact differently with the firm's financing problem. An increase in employment raises the firm's current funding requirement through the wage bill before revenues are realized. Because this additional financing need does not simultaneously expand collateral capacity, employment adjustments tend to be more closely associated with short-term finance, as in working-capital models of firm finance [Jermann and Quadrini, 2012, Mendoza, 2010]. Capital accumulation also requires current financing, but a higher choice of k_{t+1} additionally expands the firm's collateral base and the assets available for creditor recovery. It therefore relaxes the long-term borrowing constraint,

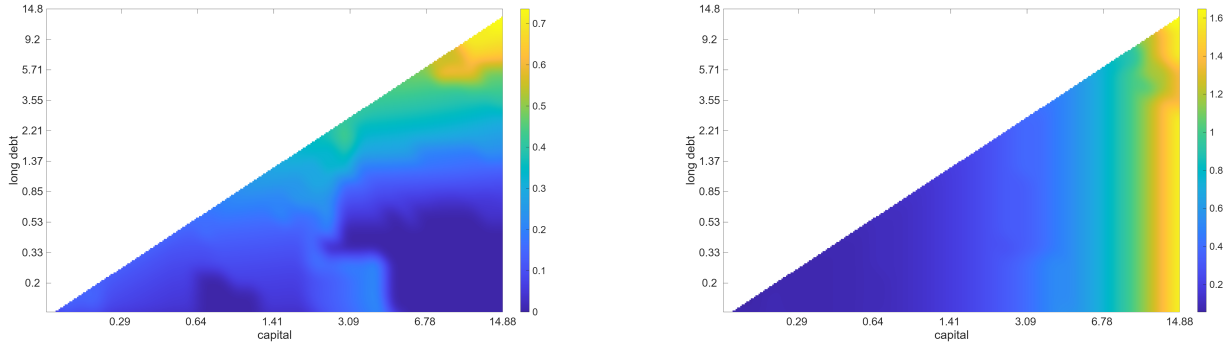
$$b_{t+1} \leq \lambda k_{t+1},$$

and improves the recovery position of short-term lenders, linking the capital decision to both sides of the firm's debt portfolio [Kiyotaki and Moore, 1997].

These differences imply that the composition of financing adjusts jointly with the firm's input choices. Employment expansions raise current funding needs without generating additional collateral, whereas capital accu-

mulation simultaneously increases financing needs and the capacity to support collateralized long-term borrowing. The resulting association between employment and short-term debt, and between capital and long-term debt, is therefore an endogenous portfolio outcome of the financing frictions rather than an imposed use-of-funds restriction.

Figure 6 illustrates the resulting relationship between debt composition and capital intensity over the firm's state space. For a given level of productivity, states with higher capital per worker tend also to be associated with a larger long-term debt share. This relationship is not mechanical: both capital intensity and debt composition are jointly determined by the firm's production and financing decisions, and depend on inherited long-term debt, collateral capacity, adjustment costs, and the endogenous price of short-term credit. The figure therefore summarizes the state-dependent interaction between firms' production structure and their debt portfolio.



(a) Long-term debt share

(b) Capital per worker

Figure 6: Debt composition and capital intensity. Panel (a) shows the long-term debt share of total debt and Panel (b) shows capital per worker, k_t/n_t, as functions of the firm's current capital and inherited long-term debt, for a given level of productivity. Policy functions are evaluated at the benchmark calibration.

The relationship in Figure 6 provides the model counterpart to the empirical association between capital intensity and debt composition documented in Section 2.5. As firms move toward more capital-intensive states, the expansion of collateral capacity makes greater use of long-term borrowing feasible and potentially attractive, while the firm continues to balance this benefit against the costs of carrying long-term debt forward. The model can therefore generate a systematic increase in the long-term debt share alongside higher capital per worker without assigning particular expenditures to particular debt instruments. The quantitative strength of this relationship is evaluated after calibration.

4.5 State Dependence and Firm Dynamics

The state-dependent financing policies described above imply that debt composition evolves systematically as firms move through the state space. As firms accumulate capital, the collateral capacity supporting long-term borrowing expands, making larger long-term debt positions feasible and potentially attractive despite the associated effects on creditor recovery and future financing conditions. More generally, the joint evolution of capital, debt, and financing capacity is a central feature of dynamic models of firms with financial frictions [Cooley and Quadrini, 2001, Clementi and Hopenhayn, 2006].

Adjustment tends to be gradual because long-term debt is costly to change and installed capital is partially irreversible. During contractions, the long-term debt stock need not fall proportionately with the firm's current

financing needs, while short-term borrowing can respond more flexibly to changes in current conditions. During expansions, by contrast, increases in collateral capacity allow firms to tilt their portfolios toward long-term finance without requiring an equally large increase in short-term borrowing. The evolution of debt composition therefore reflects both changes in firms' financing needs and the balance-sheet consequences of carrying long-term debt across periods.

These forces also have a natural life-cycle implication. Entrants begin without inherited long-term debt. As surviving firms accumulate capital, their collateral capacity expands and they can build larger long-term debt positions. At the same time, inherited long-term debt becomes increasingly costly at the margin because it reduces the recovery value available to short-term creditors and worsens future short-term financing conditions. These opposing forces generate a tendency for the long-term debt share to rise gradually over the firm life cycle rather than for firms to move immediately toward the collateral limit.

The life-cycle profile also reflects endogenous selection. Because default leads to exit, firms observed at older ages are those that have survived earlier productivity, revenue, and financing shocks. The evolution of debt composition with age therefore combines within-firm changes in capital and financing policies with selection among surviving firms. This mechanism provides a qualitative counterpart to the empirical increase in the long-term debt share with firm age documented in Section 2.4 and is consistent with evidence that larger and more mature firms tend to borrow at longer maturities [Barclay and Smith, 1995, Jungherr and Schott, 2021]. The quantitative life-cycle implications of the model are evaluated after calibration.

5 Calibration

This section describes the model calibration using annual Swedish firm-level data. We follow a two-step strategy. First, we discipline parameters with established empirical counterparts using estimates from the literature and Swedish data. Second, we jointly calibrate the remaining parameters to a set of moments that discipline firms' production, financing, survival, and debt-composition decisions. This separation makes transparent which features of the model are anchored externally and which are disciplined by the firm-level moments used in the calibration.

Importantly, the calibration does not target the central debt–factor relationships that motivate the paper. In particular, the relative associations of short- and long-term debt with employment and capital, as well as the evolution of debt composition over the firm life cycle, are left untargeted. We use these moments in Section 6to evaluate the model and the debt-portfolio mechanism developed in Section 4.

Solving the model is numerically demanding because firms' real and financial choices jointly determine the amount and price of short-term debt, the probability of default, and, following liquidation, the capital stock carried into the next period. The problem also features several non-smooth margins, including partial irreversibility of capital, fixed and variable costs of adjusting long-term debt, the collateral constraint, and transitions between repayment, liquidation, and default regions. These features make standard first-order-condition or endogenous-grid approaches difficult to apply directly. We therefore solve the Bellman problem numerically using discrete value-function iteration combined with precomputation and an off-grid refinement of the policy functions. Appendix C describes the numerical procedure in detail.

5.1 Externally Calibrated Parameters

We begin with parameters that have direct empirical counterparts or can be disciplined using estimates from the existing literature. Given that our firm-level balance-sheet data are observed annually, we set one model period to one year. We set the risk-free interest rate to $r = 0.02$. Over our 1997–2013 sample period, consistant with the annual yield on three-month Swedish Treasury securities averaged approximately 2.64 percent.[6] We set the discount factor to $\beta = 0.97$, implying an annual discount rate of approximately 3 percent, a standard value in firm-dynamics literature, [Sedláček, 2020]. Capital depreciates at an annual rate of $\delta = 0.10$, a standard value in the quantitative macroeconomic and firm-dynamics literature; see, for example, Bartelsman et al. [2013] and Devereux and Yetman [2010].

The idiosyncratic productivity process is disciplined using Swedish firm-level evidence from Carlsson et al. [2016]. We set its persistence to $\rho = 0.88$ and the standard deviation of innovations to $\sigma_\varepsilon = 0.092$. These estimates determine the persistence and dispersion of productivity shocks faced by firms.

We next discipline the parameters governing capital adjustment and borrowing frictions. We set the liquid share of capital to $\Omega = 0.01$, consistent with the range used by Khan and Thomas [2008], and the resale loss to $\nu = 0.30$, following Bloom [2009]. The collateral parameter is set to $\lambda = 0.90$, in line with with Midrigan and Xu [2014]. We set the short-term enforcement parameter to $\zeta = 0.70$, within the plaussible range of commonly used values in the literature [Liu and Wang, 2014, Rampini and Viswanathan, 2013]. The upper bound of the revenue shock is normalized to $\phi_m = 1$. Table 2 summarizes the parametrization.

[6] The series is "Interest Rates: 3-Month or 90-Day Rates and Yields: Treasury Securities: Total for Sweden," OECD Main Economic Indicators, retrieved from FRED, Federal Reserve Bank of St. Louis, series IR3TTS01SEA156N: `https://fred.stlouisfed.org/series/IR3TTS01SEA156N`. The reported 2.64 percent is the arithmetic average of the annual observations for 1997–2013.

Table 2: Parametrization

Parameter	Definition	Value	
I. Externally calibrated			
r	Risk-free interest rate	0.02	Swedish Treasury yield, 1997–2013
β	Discount factor	0.97	Sedláček [2020]
δ	Depreciation rate	0.10	Bartelsman et al. [2013]
ρ	Persistence of productivity	0.88	Carlsson et al. [2016]
σ_ε	S.D. of productivity innovations	0.092	Carlsson et al. [2016]
Ω	Liquid share of capital	0.01	Khan and Thomas [2008]
ν	Resale loss of capital	0.30	Bloom [2009]
λ	Collateral parameter	0.90	Midrigan and Xu [2014]
ζ	Short-term enforcement parameter	0.70	Liu and Wang [2014], Rampini and Viswanathan [2013]
ϕ_m	Upper bound of revenue shock	1.00	Normalization
II. Internally calibrated			
c_{ld}	Fixed cost of long-term debt	0.009	
c_f	Fixed operating cost	0.041	
c	Convex capital-adjustment cost	3.79	
ϕ_l	Lower bound of revenue shock	0.149	
α	Labor elasticity of production	0.665	
γ	Capital elasticity of production	0.186	
τ	Long-term debt adjustment cost	0.20	

5.2 Internally Calibrated Parameters

In the second step, we use the Simulated Method of Moments (SMM) to jointly calibrate the seven parameters listed in Part II of Table 2. We choose $\{c_{ld}, c_f, c, \phi_l, \alpha, \gamma, \tau\}$ to minimize the distance between a set of moments from the Swedish firm-level data and their model counterparts. The targeted moments discipline firms' factor use, overall indebtedness, participation in long-term debt markets, survival, and the relative price of short- and long-term finance. Because the parameters jointly affect firms' production and financing decisions, the mapping from individual parameters to moments is not one-to-one. We therefore interpret the moments below as being particularly informative about specific parameters rather than as providing separate identification of each parameter.

Table 3 reports the seven moments used in the internal calibration together with their model counterparts. The targets span both the production and financing margins of the model and are chosen to discipline distinct features of firms' real and financial decisions.

Table 3: SMM Results

Moment	Model	Data
Average price of short- to long-term debt	1.04	1.04
Labor-to-sales ratio	0.24	0.20
Capital-to-sales ratio	0.34	0.31
Aggregate debt-to-capital ratio	1.43	1.47
Average debt-to-sales ratio	0.49	0.46
Share of new firms surviving after 5 years	0.79	0.74
Share of firms with both debt instruments	0.73	0.68

Although all seven parameters jointly affect the targeted moments, each target is particularly informative about one parameter. The lower bound of the revenue-shock distribution, ϕ_l, is primarily disciplined by the five-year survival rate of entrants. A lower value of ϕ_l increases the probability of sufficiently adverse revenue realizations and therefore raises default and exit. The fixed cost of long-term debt, c_{ld}, is primarily disciplined by the share of firms that use both debt instruments, since a higher participation cost makes maintaining a positive long-term debt position less attractive.

The production elasticities α and γ are disciplined primarily by the labor-to-sales and capital-to-sales ratios, respectively. A higher α raises firms' desired labor input relative to output, while a higher γ raises desired capital relative to output. The fixed operating cost, c_f, is primarily informed by the average debt-to-sales ratio. A larger operating cost raises firms' financing needs for a given level of sales and therefore increases their reliance on external finance.

The convex capital-adjustment cost, c, is primarily disciplined by the aggregate debt-to-capital ratio. By governing the cost of changing the capital stock, c affects the amount of capital supporting firms' borrowing positions and hence the relationship between aggregate debt and capital. Finally, the long-term debt adjustment cost, τ, is particularly informative about the average price of short-term relative to long-term debt. A higher value of τ makes long-term debt more costly to adjust, altering firms' substitution between the two debt instruments and, through the resulting exposure to defaultable short-term borrowing, the average short-term debt price.

6 Debt Portfolios and Factor Adjustment

This section compares the relationship between debt adjustment and factor adjustment in the data with the corresponding implications of the calibrated model. We first examine whether the model reproduces the central asymmetry documented in Section 2: short-term debt is more closely associated with employment, whereas long-term debt is more closely associated with capital. We consider this relationship both in the cross section and in firms' year-to-year adjustment. We then turn to the evolution of debt composition over the firm life cycle and to its relationship with capital intensity. None of these relationships is targeted in the calibration. They therefore provide an independent dimension along which to evaluate whether the model reproduces the distinct financing patterns observed in the data.

6.1 Debt Composition across Real Margins

We begin by comparing the relationship between debt composition and firms' real inputs in the model and the data. Correlations in log levels capture how closely firms' debt positions are associated with their employment and capital stocks in the cross section. Correlations in symmetric growth rates instead capture the extent to which changes in each component of debt are associated with changes in employment and capital from one year to the next. The latter therefore speak more directly to the adjustment patterns documented in Section 2.3.

Table 4 reports both sets of moments. The model reproduces the same ordering observed in the data. In levels, short-term debt is more strongly associated with employment than long-term debt, whereas long-term debt is more strongly associated with capital than short-term debt. The same asymmetry appears in changes. Short-term debt growth is much more strongly correlated with employment growth than long-term debt growth, while capital growth is more strongly correlated with long-term debt growth than with short-term debt growth.

Table 4: Debt Composition and Factor Inputs

	Log levels		Symmetric growth rates	
Moment	Model	Data	Model	Data
Short-term debt and employment	0.98	0.61	0.92	0.11
Long-term debt and employment	0.73	0.47	0.13	0.05
Short-term debt and capital	0.81	0.50	0.36	0.09
Long-term debt and capital	0.88	0.65	0.52	0.22

Notes: Untargeted moments. The first two columns report correlations between variables expressed in logs. The last two columns report correlations between symmetric growth rates as defined in equation (1).

The mechanism behind these patterns follows from the different roles played by the two debt instruments in the firm's problem. The debt–factor asymmetry does not arise from any restriction on the use of funds. Firms are free to use either short- or long-term debt to finance any of their expenditures, and the composition of borrowing is chosen jointly with employment and capital. Instead, the asymmetry emerges because the two debt instruments differ in how flexibly they can respond to current financing needs. Short-term debt can be adjusted within the period without the fixed participation cost, collateral requirement, or adjustment cost associated with long-term borrowing. Although greater reliance on short-term debt raises repayment and default risk, it therefore provides a comparatively flexible margin for accommodating changes in current expenditures. This feature is particularly relevant for employment. Labor can be adjusted without the capital-adjustment frictions faced by investment, while the wage bill must be financed before current revenues are realized. As employment changes, firms therefore tend to accommodate a relatively larger share of the associated change in financing needs through short-term borrowing. This working-capital channel generates the stronger comovement between employment and short-term debt without imposing that wages must be financed by short-term debt.

Capital accumulation generates a different financing pattern. As firms grow, they accumulate capital and thereby expand the collateral base against which they can issue long-term debt through $b_{t+1} \leq \lambda k_{t+1}$. Conditional on having sufficient collateral, long-term debt is an attractive source of finance because it is priced at the risk-free rate, whereas short-term borrowing carries an endogenous premium reflecting repayment and default risk. At the same time, long-term debt is costly to adjust, so firms do not use it as flexibly in response to temporary

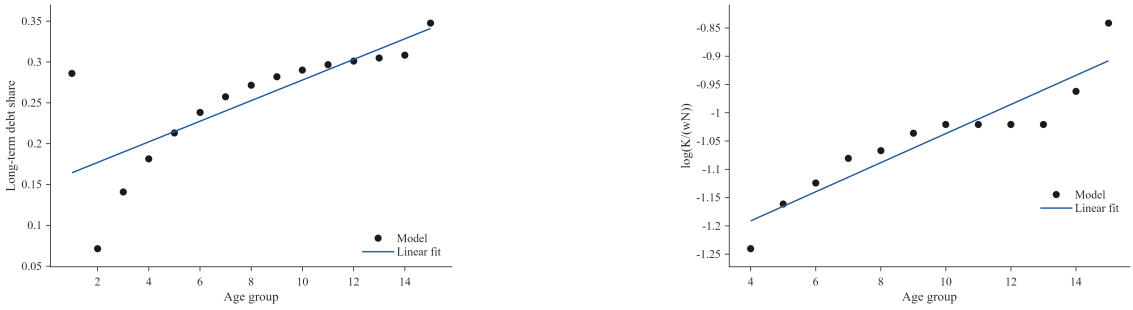
financing needs. Instead, they rely more heavily on it when their capital stock is sufficiently large to support a persistent long-term debt position. As firms accumulate capital, therefore, they increase both their capacity and their incentive to borrow long term. This mechanism generates a particularly close association between capital and long-term debt, even though the model imposes no restriction requiring capital expenditures to be financed with long-term borrowing.

Quantitatively, the model captures the ordering more closely for the capital margin than for employment adjustment. The correlation between long-term debt growth and capital growth is 0.206 in the model, compared with 0.22 in the data, while the corresponding short-term-debt correlation is smaller in both. For employment, the model also reproduces the observed ranking, but the correlation between short-term debt growth and employment growth is substantially larger in the model than in the data.

The stronger model-implied correlations are not surprising given the more parsimonious sources of heterogeneity in the model. In the data, firms differ along many observed dimensions—including industry, age, size, and balance-sheet structure—as well as along dimensions that are not directly observable, such as financing relationships, organizational structure, and other firm-specific disturbances. These sources of unobserved heterogeneity generate variation in employment, capital, and debt that need not be tightly linked to the particular financing mechanisms emphasized in the model. By contrast, the model concentrates firm heterogeneity in a smaller set of state variables and shocks, so movements in debt and factor inputs are more directly connected through the model's financing constraints and debt-portfolio choice. This tends to produce stronger correlations than in the data even when the model captures the relevant economic ordering. We therefore place greater weight on the model's ability to reproduce the relative strength of the short-debt– employment and long-debt–capital relationships than on matching each correlation coefficient exactly.

6.2 Debt Composition over the Firm Life Cycle

We next turn to the life-cycle implications of the model. As documented in Section 2.4, older firms in the data are both more capital intensive and more reliant on long-term debt. Figure 7 shows that the model reproduces the same qualitative pattern: capital per worker rises with firm age, while the share of long-term debt in firms' debt portfolios increases alongside it.



(a) Long-term debt share

(b) Capital per worker

Figure 7: Debt composition and capital intensity over the firm life cycle

Notes: Panel (a) reports the long-term debt share across firm age groups, and Panel (b) reports capital per worker across firm age groups. The points represent the model-implied age-group averages and the solid lines show the corresponding linear fits.

The life-cycle profiles reflect both selection and the endogenous evolution of surviving firms. Firms observed at

older ages are those that have survived the productivity, revenue, and financing risks encountered earlier in their lives. Selection therefore favors firms with histories of relatively favorable productivity realizations—which are persistent in the model—and revenue realizations that allow them to remain solvent. These firms tend to grow, accumulate capital, and expand their scale of operation. As their capital stocks increase, so does the collateral base against which they can issue long-term debt. Growing firms consequently acquire both a greater capacity and a stronger incentive to tilt their debt portfolios toward long-term borrowing.

This shift toward long-term debt does not come without a cost. Long-term borrowing is priced at the risk-free rate and allows the firm to reduce its current reliance on defaultable short-term credit, but a larger long-term debt position is carried into the future. By encumbering a larger share of the firm's assets, it reduces the recovery value available to future short-term creditors, raises the price of future short-term borrowing, and increases the firm's exposure to future financial distress when a low revenuw is realized. The increasing long-term debt share over the life cycle therefore reflects an optimal portfolio response to the expansion of collateral capacity, rather than an unconstrained movement toward long-term finance. As firms grow and accumulate capital, the current benefit from using relatively cheaper long-term debt becomes sufficiently large to justify the associated deterioration in future short-term financing conditions.

Adjustment frictions reinforce the persistence of this life-cycle pattern. Installed capital is partially irreversible, while long-term debt is itself costly to adjust. A negative productivity shock therefore does not induce an immediate proportional contraction in either the capital stock or the long-term debt position. In particular, partial irreversibility slows the reduction in capital and hence preserves the firm's collateral capacity even when current conditions deteriorate. Employment is considerably more flexible: a decline in productivity can lead the firm to reduce employment immediately, lowering the wage bill and the associated need for within-period finance. Short-term borrowing can therefore contract relatively quickly, while capital and long-term debt adjust more gradually. These asymmetric adjustment margins help sustain the increase in capital intensity and the long-term debt share with firm age displayed in Figure 7.

6.3 Capital Intensity and Debt Composition

The life-cycle patterns above suggest a close relationship between firms' production structure and their debt portfolios. In the data, firms with higher capital per worker also hold a larger share of their debt in long-term form, even after removing firm and time effects, as documented in Section 2.5. Figure 8 shows that the calibrated model generates the same positive relationship between capital intensity and the long-term debt share.

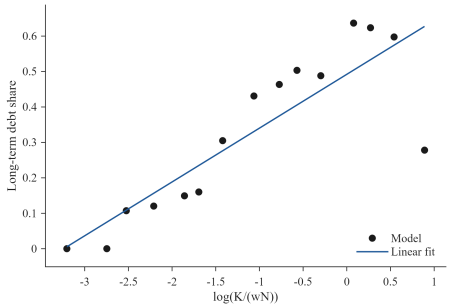


Figure 8: Capital intensity and the long-term debt share

Notes: The figure plots the model-implied relationship between capital per worker, k/n, and the share of long-term debt in total debt. Each point represents an average across firms within a capital-intensity group, and the solid line shows the corresponding linear fit.

The positive relationship between capital intensity and long-term debt reflects a trade-off between borrowing capacity and the value of preserving financial slack. Firms with larger capital stocks can support greater long-term borrowing through the collateral constraint and therefore tend to carry more long-term debt. At the same time, a larger productive scale raises the continuation value at stake when an adverse revenue realization forces the firm to liquidate capital in order to remain solvent. Pledging more of the firm's capital to long-term creditors reduces the assets available to absorb such shocks and can therefore increase the expected cost of financial distress. Firms consequently do not exhaust their long-term borrowing capacity: as capital expands they increase long-term borrowing, but retain some collateral capacity as a buffer against revenue risk. The observed increase in the long-term debt share with capital intensity reflects the balance between these two forces.

Firm growth also raises employment and, with it, the wage bill that must be financed before revenues are realized. A larger operating scale therefore leaves a greater amount of pre-financed expenditure exposed to adverse revenue realizations. Because employment does not expand the firm's collateral base and can adjust relatively quickly with current productivity and operating conditions, changes in labor demand translate more directly into changes in current financing needs. Short-term borrowing provides the more flexible margin for accommodating these needs. Thus, as firms become larger they tend to use more of both debt instruments, but the two dimensions of scale remain associated differently with the debt portfolio: capital accumulation expands the capacity for long-term borrowing, whereas higher employment increases the need for short-term finance.

Capital intensity is therefore informative about debt composition even apart from firm age. Two firms of similar scale or maturity can choose different debt portfolios because differences in their capital relative to employment change both their collateral capacity and their reliance on within-period finance. The positive relationship in Figure 8 thus provides a cross-sectional counterpart to the life-cycle mechanism: as the production structure becomes more capital intensive, the optimal debt portfolio shifts toward long-term borrowing.

7 Policy Experiment: A Guarantee on Short-Term Debt

The model provides a natural environment for studying policies that affect the relative cost of different sources of corporate finance. We focus on a policy that directly targets the friction underlying risky short-term credit: a government guarantee that raises the recovery of short-term creditors when a borrowing firm defaults. This experiment is particularly informative in our setting because the price of short-term debt is endogenous and depends on both the probability of default and the value recovered by creditors in default.

The policy can be interpreted as a partial government guarantee of short-term corporate debt. Let $\chi \geq 0$ denote the maximum additional recovery provided by the government to short-term creditors following default. In the baseline model, creditor recovery is given by equation (17). Under the policy, recovery becomes

$$\mathcal{R}_t^{\chi} = \min\left\{s_t,\, \zeta f^s(k_{t+1}, b_{t+1}) + \chi\right\}. \tag{49}$$

Thus, the guarantee supplements the private liquidation value available to short-term creditors but does not allow them to recover more than the principal of the loan. Competitive short-term creditors continue to satisfy a zero-profit condition. Replacing baseline recovery in equation (20) by (49) gives

$$s_t = \mathcal{R}_t^{\chi} G(\phi_t^e) + p_t s_t \left[1 - G(\phi_t^e)\right]. \tag{50}$$

For a given set of firm choices, a larger χ raises creditor recovery in default and therefore lowers the repayment factor required to break even. If

$$s_t \leq \zeta f^s(k_{t+1}, b_{t+1}) + \chi,$$

short-term creditors recover the full principal even following default and the competitive repayment factor is $p_t = 1$. Importantly, the guarantee is paid to creditors only after the firm has defaulted. It therefore does not itself provide the firm with additional resources with which to avoid default. Conditional on the firm's financing choices and before accounting for the tax used to finance the policy, the default threshold remains

$$\phi_t^e = \frac{p_t s_t - f^s(k_{t+1}, b_{t+1})}{y_t}, \tag{51}$$

as in equation (18). The guarantee nevertheless affects default indirectly because it changes the endogenous repayment factor p_t and induces firms to adjust their employment, capital, and debt-portfolio choices. Conditional on default, the government's payment is therefore

$$\mathcal{C}_t^{\chi} = \min\left\{\chi,\, \max\left[s_t - \zeta f^s(k_{t+1}, b_{t+1}), 0\right]\right\}. \tag{52}$$

Expected fiscal expenditure consequently depends on both the size of the guarantee and the endogenous probability of default.

Financing the guarantee. We implement the policy under a balanced government budget. The guarantee is financed by a tax on firms, paid out of resources that would otherwise accrue to shareholders. For a realization ϕ of the revenue shock, the statutory tax base is

$$\text{base}_t(\phi) = \phi y_t - p_t s_t + \max\left\{k_{t+1} - (1-\delta)k_t,\, 0\right\} + \max\left\{b_t - b_{t+1},\, 0\right\}. \tag{53}$$

Thus, capital accumulation above replacement of depreciation and reductions in long-term debt enter the tax base. Replacing depreciated capital, $k_{t+1} = (1-\delta)k_t$, does not add to the tax base, nor does a reduction in capital. Similarly, issuing additional long-term debt is neither taxed nor deducted, whereas paying down long-term debt increases the tax base.

Let τ denote the tax rate. The corresponding after-tax dividend, when the firm has sufficient cash to meet

its obligations, is

$$\widetilde{d}_t^{\tau}(\phi) = (1-\tau)\left(\phi y_t - p_t s_t\right) - \tau\left[\max\left\{k_{t+1} - (1-\delta)k_t,\, 0\right\} + \max\left\{b_t - b_{t+1},\, 0\right\}\right].$$

The firm cannot, however, pay more tax than the cash available after meeting its short-term repayment. The actual tax payment is therefore

$$T_t(\phi) = \min\left\{\max\left\{\phi y_t - p_t s_t,\, 0\right\}, \max\left\{\tau\, \text{base}_t(\phi),\, 0\right\}\right\}. \tag{54}$$

A firm with no cash after short-term repayment therefore pays no tax.

Although the guarantee itself is paid only after default, the tax used to finance it can affect firms' within-period liquidity. In particular, capital accumulation above replacement and reductions in long-term debt create an additional tax liability. A firm avoids liquidation only if its realized revenue is sufficient to cover both its short-term repayment and the tax associated with these additional components of the tax base:

$$\phi y_t \geq p_t s_t + \frac{\tau}{1-\tau}\left[\max\left\{k_{t+1} - (1-\delta)k_t,\, 0\right\} + \max\left\{b_t - b_{t+1},\, 0\right\}\right]. \tag{55}$$

The tax can therefore push a firm that would otherwise repay without liquidation into the liquidation region. By contrast, when realized revenue is insufficient to meet the short-term obligation, the firm's available cash for tax purposes is zero, so the tax does not directly reduce the resources available to avoid default.

The tax rate τ is chosen to balance the government budget in the stationary equilibrium. Let μ denote the stationary distribution of firms. Expected tax revenue from surviving firms must equal expected payments under the guarantee:

$$\int \left[\mathbb{E}\left[T_t(\phi)\right]\Pr(\text{survive} \mid x_t) - G(\phi_t^e)\mathcal{C}_t^{\chi}\right] d\mu = 0. \tag{56}$$

The equilibrium tax rate therefore increases endogenously with the fiscal resources required to support the guarantee.

The policy generates two opposing forces that are central to the experiment. First, by improving recovery in default, the guarantee reduces the price of short-term finance and makes short-term borrowing more attractive relative to collateralized long-term debt. Second, financing the guarantee through the firm tax and the endogenous response of firms' production and financing choices alter the size and composition of their debt portfolios, capital accumulation, and exposure to financial distress. The equilibrium effect of the policy therefore cannot be inferred from the direct reduction in short-term borrowing costs alone.

To compare policy intensities on an economically meaningful scale, we report the results against the fiscal cost of the guarantee as a share of aggregate sales rather than against the parameter χ itself. This normalization facilitates interpretation of the size of the intervention and separates the economic effects of the policy from the arbitrary units in which the guarantee parameter is measured.

7.1 Aggregate and Debt-Portfolio Effects

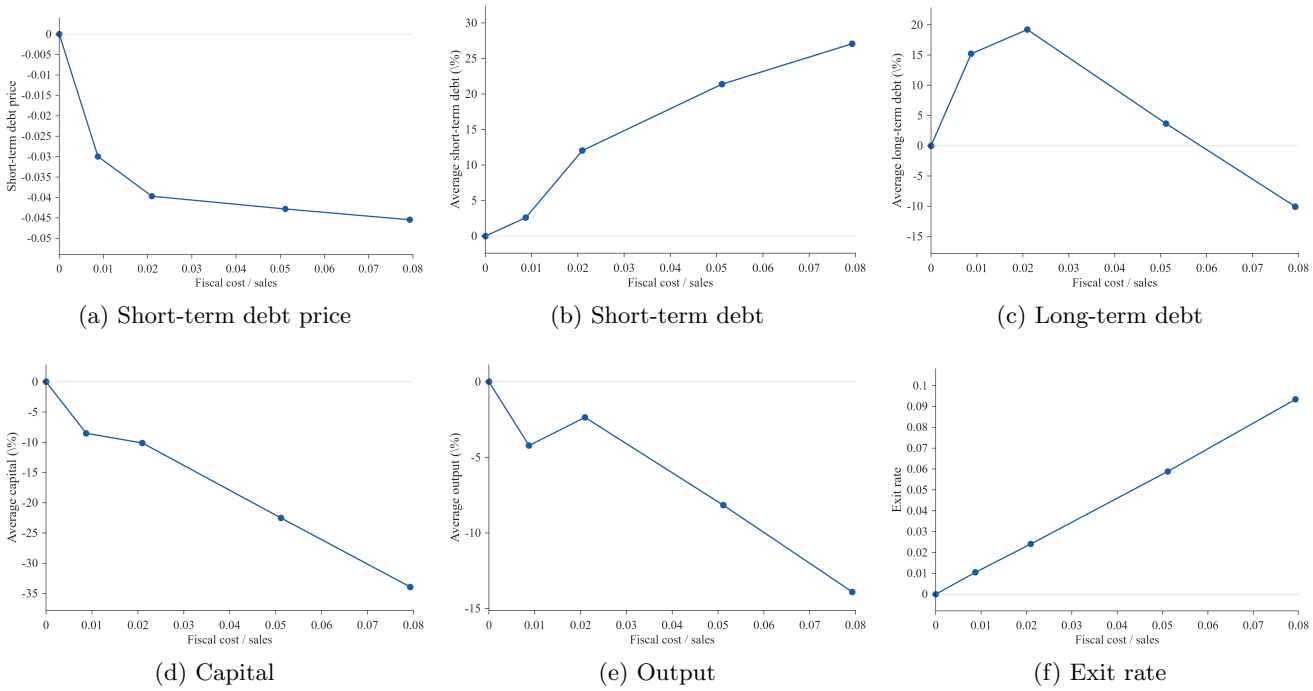
Figure 9 reports the equilibrium effects of the guarantee against its fiscal cost as a share of aggregate sales. The first effect is the direct one implied by the lenders' zero-profit condition. By increasing recovery in default, the guarantee reduces the compensation required by short-term creditors. Accordingly, the average short-term debt price falls sharply in Panel a. Most of this decline occurs at relatively small fiscal interventions, with further

increases in fiscal cost generating progressively smaller reductions in the price of short-term finance.

Firms respond strongly to the improvement in short-term borrowing terms. As shown in Panel b, short-term borrowing rises monotonically with the fiscal scale of the policy. The response of long-term debt in Panel c is instead non-monotonic, displaying a pronounced inverted-U shape. At relatively small interventions, cheaper short-term finance relaxes firms' overall financing pressure and allows them to operate at a larger desired scale. This creates room for firms to expand long-term borrowing as well, so that both components of the debt portfolio initially increase. As the guarantee becomes larger, however, short-term finance becomes increasingly attractive relative to collateralized long-term debt. Firms consequently tilt their portfolios toward the instrument directly supported by the policy, and long-term borrowing eventually declines. The non-monotonic response therefore reflects the interaction between an initial expansion in firms' financing capacity and a subsequent substitution in the composition of borrowing toward short-term debt.

This financing response also affects firms' demand for labor. The initial increase in long-term borrowing provides additional resources with which firms can support production and raises desired labor demand. Because aggregate labor supply is inelastic in the policy experiment, however, aggregate employment cannot increase. Instead, the additional demand for labor is absorbed by a higher equilibrium wage. The resulting increase in labor costs changes firms' optimal real choices and reduces the attractiveness of capital accumulation. Panel d therefore shows a decline in aggregate capital as the fiscal scale of the policy increases.

The decline in capital is reflected in aggregate production. As shown in Panel e, output falls substantially as the policy becomes larger. With aggregate labor fixed by assumption, the reduction in capital lowers the amount of productive capital available per worker and therefore reduces aggregate output. Thus, although the guarantee lowers the price of short-term finance and expands firms' use of that instrument, the general-equilibrium response of wages and capital implies that cheaper short-term credit does not translate into higher aggregate production.



(a) Short-term debt price (b) Short-term debt (c) Long-term debt

(d) Capital (e) Output (f) Exit rate

Figure 9: Default guarantees, debt portfolios, and aggregate outcomes. The figure reports the equilibrium effects of increasing the government guarantee to short-term creditors. The horizontal axis in each panel is the fiscal cost of the policy as a share of aggregate sales. Panel (a) reports the average short-term debt price, Panels (b) and (c) report short- and long-term debt, respectively, Panels (d) and (e) report aggregate capital and output, and Panel (f) reports the exit rate.

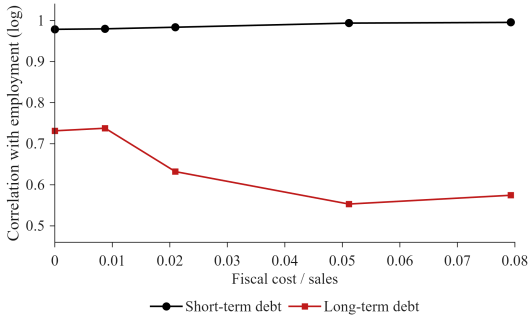
The effect on firm survival reinforces the distinction between cheaper short-term credit and a general relaxation of financial distress. As shown in Panel f, the equilibrium exit rate rises steadily with the fiscal size of the guarantee. The guarantee reduces the price charged by short-term lenders, but firms respond by taking on more short-term debt and by changing their long-term borrowing and real choices. The resulting portfolio adjustment leaves firms more exposed to the revenue realizations that determine repayment and exit.

Taken together, the results show that the policy operates through two distinct margins. At small fiscal interventions, the reduction in short-term borrowing costs relaxes firms' financing pressure sufficiently that both short- and long-term borrowing can expand. At larger interventions, the relative-price effect dominates and firms increasingly substitute toward the guaranteed short-term instrument. This reshaping of the debt portfolio is accompanied by a higher equilibrium wage, lower capital accumulation, lower output, and a higher exit rate. The experiment therefore illustrates why the composition of corporate finance matters for the evaluation of credit-support policies: making one source of finance cheaper can alter both the scale and the maturity composition of borrowing, with equilibrium effects on firms' real decisions that differ from those of a uniform relaxation of financing constraints.

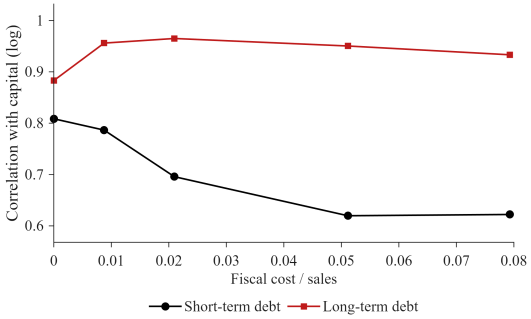
7.2 The Debt–Factor Asymmetry under the Guarantee

The portfolio reallocation documented in Section 7.1 also changes the way the two debt instruments co-move with firms' real choices. The key mechanism is that the guarantee selectively lowers the cost of short-term finance, while long-term debt remains tied to firms' collateral capacity. The policy therefore alters the relative use of

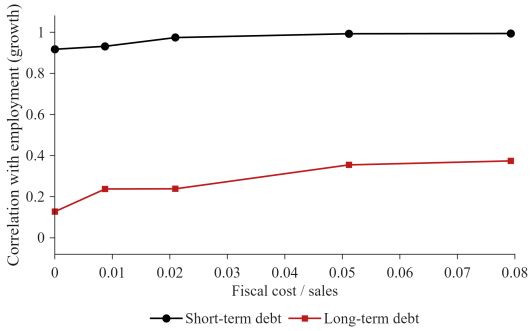
the two instruments without removing the distinction between their economic roles. Figure 10 shows how this mechanism affects the debt–factor relationships documented in Section 2.



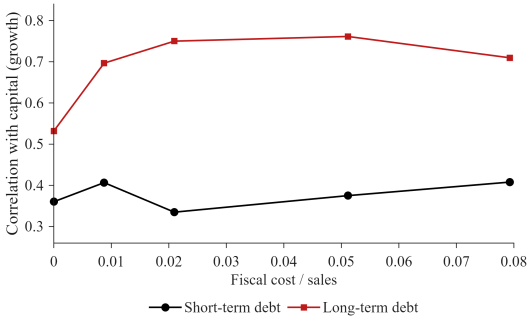
(a) Debt and employment: levels



(b) Debt and capital: levels



(c) Debt and employment: SGR



(d) Debt and capital: SGR

Figure 10: Default guarantees and the debt–factor asymmetry. The figure reports how the relationship between firms' debt portfolios and their real inputs changes with the government guarantee to short-term creditors. The horizontal axis in each panel is the fiscal cost of the guarantee as a share of aggregate sales. Panels (a) and (b) report the correlations of short- and long-term debt with employment and capital, respectively, in levels. Panels (c) and (d) report the corresponding correlations using symmetric growth rates.

In levels, the policy sharpens the separation between the two financing margins. As short-term finance becomes cheaper and firms rely more heavily on it, short-term debt remains tightly linked to employment, while its cross-sectional association with capital weakens. At the same time, the higher equilibrium wage documented in Section 7.1 weakens the association between long-term debt and employment and between short-term debt and capital. Long-term borrowing remains tied to collateral capacity and is therefore concentrated among firms with larger capital stocks. The correlation between long-term debt and capital consequently rises, while the corresponding relationship between long-term debt and employment falls. The policy thus reinforces the distinction between short-term finance as the margin associated with current operating needs and long-term finance as the margin supported by productive capital.

The same mechanism is visible in adjustment rates. Employment changes alter the wage bill immediately and therefore continue to be financed primarily through short-term debt; accordingly, the correlation between employment growth and short-term debt growth remains much stronger than the corresponding correlation with long-term debt. Capital adjustment, by contrast, changes the collateral base that supports long-term borrowing. The initial expansion in long-term finance generated by the improvement in overall financing conditions therefore strengthens the co-movement between capital growth and long-term debt growth. Even though the guarantee

makes short-term borrowing substantially more attractive in aggregate, marginal capital adjustment remains much more closely associated with the long-term financing margin.

The policy experiment therefore highlights an important feature of the model: changing the relative price of the two debt instruments changes the scale and composition of borrowing, but does not make the instruments substitutes in their relationship with firms' real decisions. The guarantee increases firms' reliance on short-term finance, while the employment–short-debt and capital–long-debt relationships remain intact and, along several dimensions, become stronger.

8 Conclusion

This paper documents a systematic relationship between the maturity composition of corporate debt and the composition of firms' real adjustment. Using Swedish firm-level data, we show that employment adjustment is substantially more closely associated with short-term debt, whereas capital adjustment is more closely associated with long-term debt. The same distinction appears in firms' longer-run evolution: as firms age and become more capital intensive, they rely relatively more on long-term borrowing, and the relationship between capital intensity and debt composition remains visible within firms. These findings suggest that debt maturity is not only a feature of firms' financial structure, but is closely connected to how their production structure evolves.

We develop a dynamic model in which employment, capital, and the debt portfolio are jointly determined. The central mechanism is an intertemporal trade-off between the two financing margins. Greater reliance on short-term debt raises current repayment and default exposure but leaves less long-term debt to be carried forward. Greater long-term borrowing reduces current reliance on risky short-term finance, but encumbers collateral and worsens the recovery position of future short-term creditors. Employment and capital interact differently with this trade-off because employment primarily changes current financing needs, whereas capital both supports production and expands collateral capacity. A calibrated version of the model reproduces the debt–factor asymmetry in both levels and adjustment rates, as well as the joint evolution of capital intensity and debt composition over the firm life cycle, despite these relationships not being targeted in the calibration.

The policy experiment illustrates why this distinction matters beyond the positive analysis. A guarantee that increases short-term creditor recovery reduces the price of short-term finance and induces firms to shift their debt portfolios toward the supported instrument. The resulting equilibrium response is not equivalent to a uniform relaxation of financing constraints: the policy raises wages, reduces capital accumulation and output, increases exit, and strengthens several dimensions of the debt–factor asymmetry. More generally, the results suggest that policies affecting corporate credit should be evaluated not only in terms of how much finance they provide, but also in terms of which part of firms' debt portfolios they alter.

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Appendices

A Data, Measurement, and Empirical Robustness

This appendix reports the robustness exercises summarized in Section 2.6. We examine whether the empirical relationships documented in Section 2 are sensitive to sample composition, the inclusion of additional balance-sheet variables, alternative fixed-effect specifications, and alternative measures of firms' real activity. Unless otherwise noted, the variables are expressed as symmetric growth rates as defined in equation (1).

The results reinforce the central empirical ordering. Employment adjustment remains more closely associated with short-term debt than with long-term debt, while capital adjustment remains more closely associated with long-term debt. The employment result is also present across a broad set of subsamples, and both relationships are robust to richer controls and specifications that absorb persistent firm heterogeneity and common variation across sectors and time.

A.1 Sample Splits

We first examine whether the relationship between employment adjustment and debt composition is concentrated among particular types of firms or periods. We re-estimate the employment specification across complementary subsamples defined by firm size, age, sector, capital intensity, and the 2008–2010 financial-crisis period.

The first set of results considers small firms, young firms, manufacturing firms, firms with capital below the sample median, and the corresponding crisis-period subsample. In every case, the estimated coefficient on short-term debt is substantially larger than the coefficient on long-term debt. The short-debt–employment relationship is therefore not driven by the largest or most mature firms, nor by firms with particularly high capital stocks.

The complementary samples deliver the same ordering. Among larger firms, older firms, non-manufacturing firms, firms with capital at or above the median, and years outside 2008–2010, employment continues to be more closely associated with short-term than with long-term debt. The fact that the ordering survives on both sides of these sample splits suggests that it is not generated by a single segment of the firm distribution.

A.2 Additional Balance-Sheet Controls and Alternative Debt Measures

We next examine whether the debt–factor relationships survive after conditioning more extensively on firms' balance sheets and whether distinguishing debt by maturity contains information beyond aggregate debt.

For employment, the pooled specifications show that the distinction between short- and long-term debt remains pronounced when the two debt components are included jointly. Adding transactions, cash, and equity reduces the magnitude of the short-debt coefficient but leaves the ordering intact: employment adjustment remains substantially more closely associated with short-term debt than with long-term debt.

The capital results display the opposite maturity ordering. When short- and long-term debt are included separately, capital adjustment is much more closely associated with long-term debt. This result is preserved when the specification includes transactions, cash, and equity. The comparison with total debt also illustrates why separating the two components of borrowing is informative: aggregation conceals the markedly different associations of short- and long-term debt with capital adjustment.

Table 5: Employment and debt composition: subsample robustness I

Debt association with employment subsamples 1

	(1)	(2)	(3)	(4)	(5)
	Size <= 10	Age <= 3	Manufacturing	Capital < median	NO2008-2010
VARIABLES	Employment	Employment	Employment	Employment	Employment
Short debt	0.0497**	0.0470**	0.0409**	0.0547**	0.0567**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Long debt	0.0234**	0.0269**	0.0268**	0.0227**	0.0249**
	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)
Cash	0.0144**	0.0169**	0.0079**	0.0145**	0.0139**
	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)
Equity	0.0517**	0.0717**	0.0621**	0.0535**	0.0561**
	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)
Observations	573,658	150,873	144,095	350,137	140,338
R^2	0.031	0.043	0.041	0.038	0.038

Notes: The table reports the relationship between symmetric employment growth and short- and long-term debt growth across subsamples. The specifications consider firms with at most 10 employees, firms no more than three years old, manufacturing firms, firms with capital below the sample median, and the 2008–2010 period. The regressions additionally control for cash and equity. Standard errors are reported in parentheses.

Table 6: Employment and debt composition: subsample robustness II

Debt association with employment subsamples 2

	(1)	(2)	(3)	(4)	(5)
	Size >= 11	Age >= 4	Non-manufacturing	Capital >= median	Yr outside 2008-2010
VARIABLES	Employment	Employment	Employment	Employment	Employment
Short debt	0.0383**	0.0483**	0.0498**	0.0432**	0.0469**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Long debt	0.0236**	0.0235**	0.0239**	0.0261**	0.0242**
	(0.001)	(0.000)	(0.000)	(0.001)	(0.000)
Cash	0.0104**	0.0124**	0.0144**	0.0120**	0.0128**
	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Equity	0.0687**	0.0502**	0.0572**	0.0626**	0.0588**
	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
Observations	243,626	666,411	673,189	467,147	676,946
R^2	0.046	0.033	0.033	0.031	0.033

Notes: The table reports the complementary subsample estimates for firms with at least 11 employees, firms at least four years old, non-manufacturing firms, firms with capital at or above the sample median, and years outside 2008–2010. The regressions additionally control for cash and equity. Standard errors are reported in parentheses.

Table 7: Employment, debt composition, and financial controls

Debt association with employment conditional on financial variables

	(1)	(2)	(3)
VARIABLES	Employment	Employment	Employment
Short debt		0.0520**	0.0441**
		(0.001)	(0.000)
Long debt		0.0214**	0.0197**
		(0.000)	(0.000)
Transactions			0.1828**
			(0.001)
Cash			-0.0006
			(0.000)
Equity			0.0392**
			(0.001)
Total debt	0.0759**		
	(0.001)		
Observations	1,146,962	813,533	751,566
R^2	0.025	0.023	0.112

Notes: The table compares a specification using total debt with specifications that distinguish short- and long-term debt and progressively introduce additional financial controls. All variables are expressed as symmetric growth rates. Standard errors are reported in parentheses.

A.3 Fixed Effects and Alternative Specifications

An important concern is that the baseline associations could reflect persistent differences across firms or common shocks affecting firms within the same sector and year. We therefore progressively introduce year–sector fixed effects, age controls, size controls, and firm fixed effects.

For employment, the short-term debt coefficient remains larger than the long-term debt coefficient in every specification. The result survives the inclusion of year–sector effects and observable firm characteristics and, most importantly, remains present when identification comes from variation within firms over time. Persistent differences in firms' production technologies, financing relationships, or other time-invariant characteristics therefore do not account for the empirical ordering.

The capital relationship is equally stable. Across specifications with year–sector effects, age and size controls, and firm fixed effects, the coefficient on long-term debt remains substantially larger than the coefficient on short-term debt. The maturity ordering for capital is thus not explained by persistent cross-sectional heterogeneity or common sectoral movements.

A.4 Alternative Measures of Real Activity

We finally consider alternative measures of firms' real activity. The first exercise replaces employment with the wage bill. Figure 11 shows that the time-series ordering closely resembles the employment result: changes in the wage bill are more strongly associated with short-term debt than with long-term debt, and the difference widens sharply around the financial crisis. This provides additional evidence that short-term borrowing is particularly

Table 8: Capital, debt composition, and financial controls

Debt association with assets B conditional on financial variables

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	Assets B	Assets B	Assets B	Assets B	Assets B	Assets B
Short debt		0.0616**		0.0600**	0.0603**	0.0603**
		(0.002)		(0.002)	(0.002)	(0.002)
Long debt		0.2060**	0.2073**	0.2028**	0.2007**	0.1991**
		(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Total debt	0.2756**					
	(0.003)					
Short debt and Transaction Debt			0.1893**			
			(0.003)			
Transactions				0.1245**	0.1320**	0.1275**
				(0.003)	(0.003)	(0.003)
Cash					-0.0196**	-0.0298**
					(0.001)	(0.001)
Equity						0.1278**
						(0.004)
Observations	401,224	339,444	342,747	339,274	332,882	311,629
R^2	0.062	0.072	0.080	0.078	0.079	0.081

Notes: The table compares specifications using total debt, short- and long-term debt separately, and progressively richer sets of financial variables. The dependent variable is capital adjustment. Standard errors are reported in parentheses.

Table 9: Employment and debt composition: alternative specifications

Debt association with employment conditional on financial variables robustness

	(1)	(2)	(3)	(4)	(5)
				Year-sector FE	
			Year-sector FE	Age	
		Year-sector FE	Age	Size	Firm FE
VARIABLES	Employment	Employment	Employment	Employment	Employment
Short debt	0.0438**	0.0431**	0.0428**	0.0413**	0.0340**
	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)
Long debt	0.0202**	0.0202**	0.0202**	0.0195**	0.0188**
	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)
Transactions	0.1834**	0.1820**	0.1808**	0.1757**	0.1406**
	(0.001)	(0.003)	(0.003)	(0.003)	(0.001)
Cash	0.0002	0.0002	0.0006	0.0010*	0.0009*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Equity	0.0389**	0.0387**	0.0353**	0.0339**	0.0221**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Observations	817,253	804,082	804,082	804,082	817,253
R^2	0.111	0.109	0.110	0.130	0.089
Number of YearSektor		272	272	272	
Number of firmID					152,053

Notes: The table reports specifications with progressively richer sets of controls and fixed effects. The final column includes firm fixed effects. Standard errors are reported in parentheses.

closely related to firms' current operating margin.

Value added provides a broader measure of firm activity. When total debt is decomposed into its short- and long-term components, value-added adjustment is more strongly associated with short-term debt. The result is preserved after transactions, cash, and equity are introduced, indicating that the short-term debt relationship is not simply capturing movements in these other balance-sheet items.

The relationship is also robust to the same alternative fixed-effect specifications used for employment and capital. In particular, the larger association of value added with short-term debt remains after including year–sector effects, age and size controls, and firm fixed effects.

A.5 Summary

The robustness exercises reinforce the empirical patterns emphasized in the main text. The stronger association between employment and short-term debt is present across a wide range of firm and time subsamples and survives specifications that absorb sector–year variation and persistent firm heterogeneity. The opposite ordering for capital is similarly robust to additional financial controls and alternative fixed-effect specifications. Alternative measures of current firm activity, including the wage bill and value added, also display a comparatively strong association with short-term borrowing. Taken together, these results indicate that the debt–factor asymmetry is not driven by a particular sample definition, a small set of balance-sheet controls, or persistent differences across firms.

Table 10: Capital and debt composition: alternative specifications

Debt association with assets B conditional on financial variables robustness

	(1)	(2)	(3)	(4)	(5)
				Year-sector FE	
			Year-sector FE	Age	
		Year-sector FE	Age	Size	Firm FE
VARIABLES	Assets B	Assets B	Assets B	Assets B	Assets B
Short debt	0.0603**	0.0604**	0.0593**	0.0595**	0.0508**
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Long debt	0.1991**	0.1984**	0.1964**	0.1964**	0.1910**
	(0.002)	(0.005)	(0.005)	(0.005)	(0.002)
Transactions	0.1275**	0.1268**	0.1243**	0.1252**	0.1137**
	(0.003)	(0.005)	(0.005)	(0.005)	(0.004)
Cash	-0.0298**	-0.0297**	-0.0284**	-0.0285**	-0.0241**
	(0.001)	(0.002)	(0.002)	(0.002)	(0.001)
Equity	0.1278**	0.1277**	0.1152**	0.1149**	0.0981**
	(0.004)	(0.006)	(0.006)	(0.006)	(0.005)
Observations	311,629	308,127	308,127	308,127	311,629
R^2	0.081	0.071	0.075	0.075	0.109
Number of YearSektor		272	272	272	
Number of firmID					53,727

Notes: The table reports specifications with progressively richer sets of controls and fixed effects. The final column includes firm fixed effects. Standard errors are reported in parentheses.

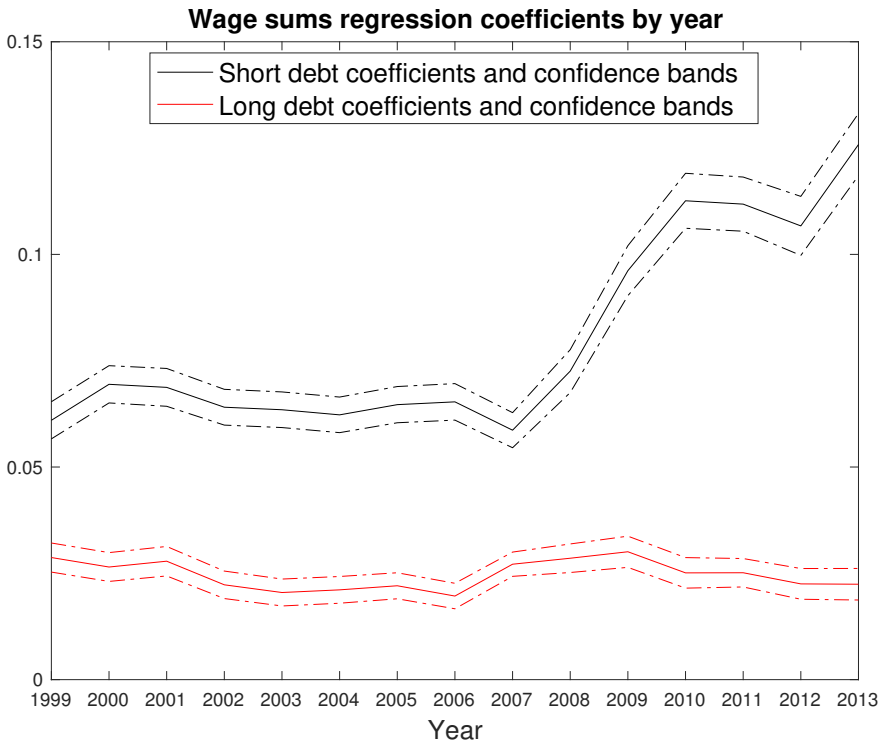


Figure 11: Debt composition and the wage bill over time

Notes: The figure reports year-specific coefficients on short- and long-term debt when the symmetric growth rate of the wage bill is the dependent variable. The dashed lines report the corresponding confidence bands.

Table 11: Value added, debt composition, and financial controls

Debt association with value added conditional on financial variables

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	Value added	Value added	Value added	Value added	Value added	Value added
Short debt		0.0725**		0.0638**	0.0623**	0.0627**
		(0.001)		(0.001)	(0.001)	(0.001)
Long debt		0.0150**	0.0178**	0.0107**	0.0148**	0.0224**
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Total debt	0.0908**					
	(0.001)					
Short debt and Transaction Debt			0.2823**			
			(0.001)			
Transactions				0.2664**	0.2520**	0.2382**
				(0.001)	(0.001)	(0.001)
Cash					0.0413**	0.0234**
					(0.000)	(0.000)
Equity						0.2367**
						(0.001)
Observations	1,259,729	882,240	891,496	881,813	864,460	817,177
R^2	0.032	0.033	0.121	0.145	0.157	0.220

Notes: The table reports specifications relating value-added adjustment to total debt and to the separate components of the debt portfolio, with progressively richer financial controls. Standard errors are reported in parentheses.

Table 12: Value added and debt composition: alternative specifications

Debt association with value added conditional on financial variables robustness

	(1)	(2)	(3)	(4)	(5)
				Year-sector FE	
			Year-sector FE	Age	
		Year-sector FE	Age	Size	Firm FE
VARIABLES	Value added	Value added	Value added	Value added	Value added
Short debt	0.0627**	0.0622**	0.0616**	0.0607**	0.0523**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Long debt	0.0224**	0.0226**	0.0225**	0.0221**	0.0210**
	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)
Transactions	0.2382**	0.2365**	0.2340**	0.2311**	0.1939**
	(0.001)	(0.003)	(0.003)	(0.003)	(0.001)
Cash	0.0234**	0.0234**	0.0241**	0.0243**	0.0239**
	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)
Equity	0.2367**	0.2359**	0.2294**	0.2286**	0.2262**
	(0.001)	(0.003)	(0.003)	(0.003)	(0.002)
Observations	817,177	804,006	804,006	804,006	817,177
R^2	0.220	0.213	0.217	0.221	0.195
Number of YearSektor		272	272	272	
Number of firmID					152,038

Notes: The table reports specifications with progressively richer sets of controls and fixed effects. The final column includes firm fixed effects. Standard errors are reported in parentheses.

B Model Details and Derivations

This appendix provides additional details on the financial-distress regions, the liquidation problem, the endogenous price of short-term credit, and the firm's optimality conditions. We focus on locally differentiable regions of the policy functions. The numerical solution itself allows for the kinks generated by partial irreversibility, the absolute-value adjustment cost for long-term debt, the fixed participation cost in long-term borrowing, the collateral constraint, and the recovery function.

For the derivations, it is convenient to use the financing identity in equation (11) to substitute for short-term borrowing. This is only an algebraic reduction of the firm's problem: employment, capital, and the composition of the debt portfolio remain jointly determined.

B.1 A Two-State Illustration of Short-Term Debt Pricing

Consider a special case in which the revenue shock has two realizations. Let π denote the probability of the high-revenue state, in which the firm repays the promised amount ps, and let $1-\pi$ denote the probability of default. Conditional on default, short-term creditors recover

$$R = \min\left\{s,\, \zeta f^s(k', b')\right\}.$$

Competitive lenders therefore satisfy

$$s = (1-\pi)R + \pi ps. \tag{57}$$

Solving for the short-term debt price gives

$$p = \frac{1}{\pi}\left[1 - (1-\pi)\frac{R}{s}\right]. \tag{58}$$

If $\zeta f^s(k', b') \geq s$, creditors recover the full principal in default, $R = s$, and equation (50) implies $p = 1$. Consider instead the partial-recovery region,

$$\zeta f^s(k', b') < s,$$

so that

$$R = \zeta f^s(k', b').$$

The price becomes

$$p = \frac{1}{\pi}\left[1 - (1-\pi)\frac{\zeta f^s(k', b')}{s}\right]. \tag{59}$$

Lemma. In the partial-recovery region, the short-term debt price satisfies

$$\frac{\partial p}{\partial \pi} < 0, \qquad \frac{\partial p}{\partial s} > 0, \qquad \frac{\partial p}{\partial f^s} < 0. \tag{60}$$

Moreover, if

$$\frac{\partial f^s}{\partial k'} > 0 \qquad \text{and} \qquad \frac{\partial f^s}{\partial b'} < 0,$$

then

$$\frac{\partial p}{\partial k'} < 0, \qquad \frac{\partial p}{\partial b'} > 0. \tag{61}$$

Proof. From equation (51),

$$\frac{\partial p}{\partial s} = \frac{1-\pi}{\pi} \frac{\zeta f^s(k', b')}{s^2} > 0,$$

and

$$\frac{\partial p}{\partial f^s} = -\frac{(1-\pi)\zeta}{\pi s} < 0.$$

Furthermore,

$$\frac{\partial p}{\partial \pi} = -\frac{1}{\pi^2}\left[1 - \frac{\zeta f^s(k', b')}{s}\right] < 0,$$

where the inequality follows from $\zeta f^s(k', b') < s$. Finally,

$$\frac{\partial p}{\partial k'} = \frac{\partial p}{\partial f^s}\frac{\partial f^s}{\partial k'} < 0,$$

while

$$\frac{\partial p}{\partial b'} = \frac{\partial p}{\partial f^s}\frac{\partial f^s}{\partial b'} > 0.$$

□

The two-state case therefore makes transparent the forces operating in the general pricing equation. A higher repayment probability lowers the price of short-term debt, while greater borrowing raises it by increasing the lender's exposure relative to available recovery. Greater liquidation capacity reduces the price by improving recovery in default. Hence, capital tends to improve short-term borrowing terms through its effect on liquidation value, whereas long-term debt tends to worsen them by reducing the capital available for liquidation.

B.2 Realized Dividends and Financial-Distress Regions

Let

$$q_t \equiv p_t s_t \tag{62}$$

denote the firm's promised short-term repayment. The repayment and default thresholds defined in equations (22) and (18) are

$$\phi_t^r = \frac{q_t}{y_t}, \tag{63}$$

and

$$\phi_t^e = \frac{q_t - f^s(k_{t+1}, b_{t+1})}{y_t}. \tag{64}$$

The firm chooses its realized dividend after the revenue shock has been observed. Limited liability requires

$$d_t(\phi) \geq 0. \tag{65}$$

For realizations $\phi \geq \phi_t^r$, the firm can repay its short-term obligation without liquidating capital. Feasibility requires

$$d_t(\phi) + q_t \leq \phi y_t. \tag{66}$$

Since distributions enter shareholder value one-for-one, the constraint binds whenever the firm is in this region:

$$d_t(\phi) = \phi y_t - q_t, \qquad \phi \geq \phi_t^r. \tag{67}$$

For

$$\phi_t^e \leq \phi < \phi_t^r,$$

current revenue is insufficient to meet the short-term obligation. The firm remains solvent by liquidating part of its chosen capital stock. Because it liquidates only the amount required to meet the repayment obligation, shareholder distributions in this region are

$$d_t(\phi) = 0, \qquad \phi_t^e \leq \phi < \phi_t^r. \tag{68}$$

For $\phi < \phi_t^e$, the firm defaults and exits, and shareholder value is given by equation (23). Hence, conditional on survival, the realized dividend policy can be summarized as

$$d_t(\phi) = \max\left\{\phi y_t - q_t,\, 0\right\}. \tag{69}$$

B.3 Capital Liquidation

Recall from equation (13) that the maximum proceeds that the firm can obtain through capital liquidation are

$$f^s(k_{t+1}, b_{t+1}) = \max_{\tilde{k}_{t+1} \geq b_{t+1}/\lambda} \mathcal{L}(k_{t+1}, \tilde{k}_{t+1}), \tag{70}$$

where $\mathcal{L}(k_{t+1}, \tilde{k}_{t+1})$ denotes net proceeds from reducing the chosen capital stock from k_{t+1} to $\tilde{k}_{t+1}$, taking account of the resale and adjustment costs specified in Section 4.2.1.

Let $\eta_t \geq 0$ denote the multiplier on the collateral requirement

$$\tilde{k}_{t+1} - \frac{b_{t+1}}{\lambda} \geq 0.$$

Away from kinks in the liquidation technology, the first-order and complementary-slackness conditions for the liquidation problem are

$$\mathcal{L}_2\left(k_{t+1}, \tilde{k}_{t+1}\right) + \eta_t = 0, \tag{71}$$

and

$$\eta_t \left(\tilde{k}_{t+1} - \frac{b_{t+1}}{\lambda}\right) = 0. \tag{72}$$

The envelope theorem therefore implies

$$f_k^s \equiv \frac{\partial f^s(k_{t+1}, b_{t+1})}{\partial k_{t+1}} = \mathcal{L}_1\left(k_{t+1}, \tilde{k}_{t+1}^*\right) \geq 0, \tag{73}$$

and

$$f_b^s \equiv \frac{\partial f^s(k_{t+1}, b_{t+1})}{\partial b_{t+1}} = -\frac{\eta_t}{\lambda} \leq 0. \tag{74}$$

Thus, additional capital weakly increases the resources available in liquidation, whereas additional long-term

debt weakly reduces them by raising the amount of capital that must remain pledged to long-term creditors.

Within the liquidation region, the firm sells only enough capital to cover the difference between its short-term obligation and current revenue. Consequently,

$$\mathcal{L}\left(k_{t+1}, \tilde{k}_{t+1}(\phi)\right) = q_t - \phi y_t, \qquad \phi_t^e \leq \phi < \phi_t^r. \tag{75}$$

At the upper boundary,

$$\tilde{k}_{t+1}(\phi_t^r) = k_{t+1},$$

while at the default threshold the required proceeds equal the maximum amount that can be raised,

$$q_t - \phi_t^e y_t = f^s(k_{t+1}, b_{t+1}).$$

For any choice variable a, differentiation of equation (67) gives, on a regular branch,

$$\frac{\partial \tilde{k}_{t+1}(\phi)}{\partial a} = \frac{q_{t,a} - \phi y_{t,a} - \mathcal{L}_1 \mathbf{1}\{a = k_{t+1}\}}{\mathcal{L}_2}, \tag{76}$$

where subscripts denote partial derivatives evaluated at the relevant policy choices.

B.4 Short-Term Credit Pricing

The recovery of short-term creditors is

$$\mathcal{R}_t = \min\left\{s_t,\, \zeta f^s(k_{t+1}, b_{t+1})\right\}. \tag{77}$$

The lenders' zero-profit condition in equation (20) can be written as

$$F(p_t, s_t, y_t, k_{t+1}, b_{t+1}) = \mathcal{R}_t G(\phi_t^e) + p_t s_t \left[1 - G(\phi_t^e)\right] - s_t = 0, \tag{78}$$

where

$$\phi_t^e = \frac{p_t s_t - f^s(k_{t+1}, b_{t+1})}{y_t}. \tag{79}$$

There are two recovery regions.

Full recovery. If

$$s_t \leq \zeta f^s(k_{t+1}, b_{t+1}),$$

then $\mathcal{R}_t = s_t$. Provided the probability of repayment is positive, equation (70) becomes

$$(p_t - 1) s_t \left[1 - G(\phi_t^e)\right] = 0,$$

and hence

$$p_t = 1. \tag{80}$$

In this region lenders recover the full principal even in default and no default premium is required.

Limited recovery. Consider instead the interior region

$$\mathcal{R}_t = \zeta f^s(k_{t+1}, b_{t+1}) < s_t. \tag{81}$$

Let

$$G_t \equiv G(\phi_t^e), \qquad g_t \equiv g(\phi_t^e).$$

The derivative of the default threshold with respect to the repayment factor is

$$\frac{\partial \phi_t^e}{\partial p_t} = \frac{s_t}{y_t}. \tag{82}$$

Differentiating the zero-profit condition with respect to p_t gives

$$F_p = s_t(1 - G_t) - \frac{s_t(p_t s_t - \mathcal{R}_t)}{y_t} g_t. \tag{83}$$

Holding p_t fixed, differentiation with respect to short-term borrowing gives

$$\begin{aligned} F_s &= p_t(1 - G_t) - 1 - \frac{p_t(p_t s_t - \mathcal{R}_t)}{y_t} g_t \\ &= -\frac{\mathcal{R}_t}{s_t} G_t - \frac{p_t(p_t s_t - \mathcal{R}_t)}{y_t} g_t < 0, \end{aligned} \tag{84}$$

where the second equality uses the zero-profit condition.

Similarly,

$$F_y = \frac{(p_t s_t - \mathcal{R}_t)\left[p_t s_t - f^s(k_{t+1}, b_{t+1})\right]}{y_t^2} g_t > 0, \tag{85}$$

$$F_k = \left[\zeta G_t + \frac{p_t s_t - \mathcal{R}_t}{y_t} g_t\right] f_k^s > 0, \tag{86}$$

and

$$F_b = \left[\zeta G_t + \frac{p_t s_t - \mathcal{R}_t}{y_t} g_t\right] f_b^s < 0. \tag{87}$$

On a locally regular branch with

$$F_p > 0, \tag{88}$$

the implicit-function theorem yields

$$H_j = -\frac{F_j}{F_p}, \qquad j \in \{s, y, k, b\}. \tag{89}$$

Equations (76)–(79) therefore imply

$$H_s > 0, \qquad H_y < 0, \qquad H_k < 0, \qquad H_b > 0, \tag{90}$$

as reported in equation (45).

B.5 Financing Choices and the Short-Term Debt Price

Using equation (34), define

$$s_t = S(x_t, n_t, k_{t+1}, b_{t+1}).$$

Away from adjustment-cost kinks and the fixed long-debt participation margin, the relevant derivatives are

$$s_{t,n} = w_t, \tag{91}$$

$$s_{t,k} = C_{k_{t+1}} - (\Delta k_t)_{k_{t+1}}, \tag{92}$$

and

$$s_{t,b} = -1 + \psi_{b_{t+1}}. \tag{93}$$

The total response of the short-term repayment factor to the firm's choices is therefore

$$p_{t,n} = H_s s_{t,n} + H_y y_{t,n}, \tag{94}$$

$$p_{t,k} = H_s s_{t,k} + H_k, \tag{95}$$

and

$$p_{t,b} = H_s s_{t,b} + H_b. \tag{96}$$

Here k and b in the derivative subscripts refer to the choices k_{t+1} and b_{t+1}, respectively. Since current output is produced with predetermined k_t,

$$\frac{\partial y_t}{\partial k_{t+1}} = \frac{\partial y_t}{\partial b_{t+1}} = 0,$$

whereas

$$y_{t,n} = \alpha \frac{y_t}{n_t}. \tag{97}$$

The promised short-term obligation is $q_t = p_t s_t$. Its derivative with respect to any firm choice a is

$$q_{t,a} = p_t s_{t,a} + s_t p_{t,a}. \tag{98}$$

Consequently, the total derivatives of the repayment and default thresholds are

$$\phi^r_{t,a} = \frac{q_{t,a} y_t - q_t y_{t,a}}{y_t^2}, \tag{99}$$

and

$$\phi^e_{t,a} = \frac{[q_{t,a} - f^s_a]\, y_t - [q_t - f^s]\, y_{t,a}}{y_t^2}, \tag{100}$$

where

$$f^s_n = 0, \qquad f^s_k = \frac{\partial f^s}{\partial k_{t+1}}, \qquad f^s_b = \frac{\partial f^s}{\partial b_{t+1}}.$$

These expressions make explicit that a production or portfolio choice affects default risk through three distinct

margins: the amount of short-term finance required, the repayment factor charged on that finance, and the resources available to the firm and its creditors after the revenue shock.

B.6 Reduced Recursive Problem

After substituting the optimal realized dividend policy and using the financing identity to eliminate s_t, the firm's recursive problem can be written as

$$V(z_t, k_t, b_t) = \max_{n_t, k_{t+1}, b_{t+1}} \Bigg\{ \int_{\phi_t^r}^{\phi_m} \left[\phi y_t - q_t + \beta \mathbb{E}_t V(z_{t+1}, k_{t+1}, b_{t+1})\right] g(\phi)\, d\phi + \int_{\phi_t^e}^{\phi_t^r} \beta \mathbb{E}_t V\left(z_{t+1}, \tilde{k}_{t+1}(\phi), b_{t+1}\right) g(\phi)\, d\phi + \beta \int_{\phi_l}^{\phi_t^e} V^e(k_{t+1}, b_{t+1}, s_t) g(\phi)\, d\phi \Bigg\}, \tag{101}$$

subject to

$$b_{t+1} \leq \lambda k_{t+1}. \tag{102}$$

For compactness, define

$$V_t^R \equiv \mathbb{E}_t V(z_{t+1}, k_{t+1}, b_{t+1}), \tag{103}$$

and

$$V_t^L(\phi) \equiv \mathbb{E}_t V\left(z_{t+1}, \tilde{k}_{t+1}(\phi), b_{t+1}\right). \tag{104}$$

At the repayment threshold,

$$\phi_t^r y_t - q_t = 0$$

and

$$\tilde{k}_{t+1}(\phi_t^r) = k_{t+1}.$$

Hence the value of the firm is continuous when moving between the full repayment and liquidation regions, and the boundary term associated with ϕ_t^r cancels when differentiating equation (93).

At the default threshold, define

$$\Gamma_t^e \equiv \beta \left[V^e(k_{t+1}, b_{t+1}, s_t) - V_t^L(\phi_t^e)\right]. \tag{105}$$

The term Γ_t^e captures the change in discounted shareholder value associated with crossing from continuation after maximal capital liquidation into default and exit.

B.7 Optimality Conditions

Let $\mu_t \geq 0$ denote the multiplier on the collateral constraint written as

$$\lambda k_{t+1} - b_{t+1} \geq 0.$$

The complementary-slackness condition is

$$\mu_t \left(\lambda k_{t+1} - b_{t+1}\right) = 0. \tag{106}$$

The following conditions apply to interior choices away from the discrete long-term debt participation margin and the kinks in the adjustment technologies.

Employment. Differentiating equation (93) with respect to employment gives

$$\begin{aligned} 0 = & \int_{\phi_t^r}^{\phi_m} \left[\phi y_{t,n} - q_{t,n}\right] g(\phi)\, d\phi \\ & + \beta \int_{\phi_t^e}^{\phi_t^r} V_{t,k}^L(\phi) \tilde{k}_{t+1,n}(\phi) g(\phi)\, d\phi \\ & + \beta G(\phi_t^e) V_n^e + \Gamma_t^e g(\phi_t^e) \phi_{t,n}^e. \end{aligned} \tag{107}$$

The first line captures the effect of employment on realized operating surplus in states in which no liquidation is required. The remaining terms capture the effect of employment on the amount of capital that must be liquidated, on shareholder value in default, and on the probability of crossing the default boundary.

In the limited-recovery region,

$$\mathcal{R}_t = \zeta f^s(k_{t+1}, b_{t+1}) < s_t,$$

equation (23) reduces to

$$V^e = (1-\zeta) f^s(k_{t+1}, b_{t+1}), \tag{108}$$

so that

$$V_n^e = 0. \tag{109}$$

Capital. For an interior choice of k_{t+1},

$$\begin{aligned} 0 = & \int_{\phi_t^r}^{\phi_m} \left[-q_{t,k} + \beta V_{t,k}^R\right] g(\phi)\, d\phi \\ & + \beta \int_{\phi_t^e}^{\phi_t^r} V_{t,k}^L(\phi) \tilde{k}_{t+1,k}(\phi) g(\phi)\, d\phi \\ & + \beta G(\phi_t^e) V_k^e + \Gamma_t^e g(\phi_t^e) \phi_{t,k}^e + \mu_t \lambda. \end{aligned} \tag{110}$$

The capital choice affects firm value through its current financing cost, future productive capacity, the amount of capital retained following liquidation, creditor recovery and default risk, and the collateral capacity supporting long-term debt.

In the limited-recovery region, equation (100) implies

$$V_k^e = (1-\zeta) f_k^s. \tag{111}$$

Long-term debt. For an interior positive choice of b_{t+1} away from the adjustment-cost kink,

$$\begin{aligned} 0 = & \int_{\phi_t^r}^{\phi_m} \left[-q_{t,b} + \beta V_{t,b}^R\right] g(\phi)\, d\phi \\ & + \beta \int_{\phi_t^e}^{\phi_t^r} \left[V_{t,k}^L(\phi)\tilde{k}_{t+1,b}(\phi) + V_{t,b}^L(\phi)\right] g(\phi)\, d\phi \\ & + \beta G(\phi_t^e) V_b^e + \Gamma_t^e g(\phi_t^e)\phi_{t,b}^e - \mu_t. \end{aligned} \tag{112}$$

In the limited-recovery region,

$$V_b^e = (1-\zeta) f_b^s. \tag{113}$$

The condition therefore balances the current funding benefit from additional long-term borrowing against its effects on the short-term repayment factor, liquidation capacity, future continuation value, default risk, and the collateral constraint.

At $b_{t+1} = 0$, the fixed participation cost $c_{ld}\mathbf{1}\{b_{t+1} > 0\}$ creates a discrete choice. The relevant condition is therefore a value comparison between not participating in the long-term debt market and choosing the optimal strictly positive long-term debt position; equation (104) applies only conditional on positive participation.

Similarly, at $b_{t+1} = b_t$ the absolute-value adjustment cost

$$\psi(b_t, b_{t+1}) = \tau|b_{t+1} - b_t|$$

is not differentiable. The optimal policy at this point is characterized by the corresponding subgradient condition rather than equation (104).

B.8 Default Value across Recovery Regions

For completeness, shareholder value in default differs across the two recovery regions.

If recovery is limited by the liquidation value available to short-term creditors,

$$\zeta f^s(k_{t+1}, b_{t+1}) < s_t,$$

then

$$V^e = (1-\zeta) f^s(k_{t+1}, b_{t+1}), \tag{114}$$

with derivatives given in equations (101)–(105).

If instead short-term creditors can recover the full principal,

$$s_t \le \zeta f^s(k_{t+1}, b_{t+1}),$$

then

$$V^e = f^s(k_{t+1}, b_{t+1}) - s_t. \tag{115}$$

In this region $p_t = 1$, but changes in the firm's real and financial choices can still affect shareholder value through the amount of principal s_t and through the liquidation value $f^s(k_{t+1}, b_{t+1})$.

The numerical solution evaluates the full piecewise problem rather than imposing the interior conditions

derived above. The analytical conditions are therefore useful for interpreting the forces governing the policy functions, while the computed policies also incorporate the discrete participation decision, binding collateral constraints, adjustment-cost kinks, and changes between full- and limited-recovery regions.

C Numerical Solution and Calibration Procedure

The numerical difficulty of the model arises from the interaction between the firm's real and financial choices and the endogenous short-term debt price. For each state and candidate choice, employment, capital, and long-term debt determine the residual short-term financing requirement, while the resulting short-term debt price depends on default risk and creditor recovery. The same choices determine the repayment and default thresholds and, in the liquidation region, the capital stock carried into the next period.

The algorithm exploits the fact that many of these objects do not depend on the value function. For every state–choice pair, we therefore precompute the current dividend, short-term debt price, regime probabilities, and the mapping from revenue shocks into post-liquidation capital. These objects are stored outside the value-function iteration. The revenue-shock distribution is approximated using twenty bins, choices that violate the collateral constraint or admit no feasible short-term debt price are removed before the iteration, and the remaining problem reduces to forming expected continuation values and maximizing over feasible choices.

The implementation combines discrete value-function iteration with an off-grid refinement of the choice variables. It exploits matrix operations in MATLAB, while the computationally intensive parts of the value-function iteration are implemented using MATLAB MEX routines compiled in C. This substantially reduces the computational burden associated with repeatedly evaluating the firm problem over the state and choice grids.

C.1 Numerical Algorithm

Productivity is approximated by a finite-state Markov chain, using either Tauchen's method or the Rouwenhorst method according to the persistence of the productivity process in equation (5). Capital, long-term debt, and labor are represented on finite grids. A state is a node

$$(z, k, b),$$

and a discrete choice is a node

$$(n, k', b').$$

For every state–choice pair, the following objects are evaluated before value-function iteration and stored as arrays.

1. Output,

   $$y = zf(n, k),$$

   using the production function in equation (4).

2. The resources generated by capital adjustment, the capital adjustment cost, and the long-term-debt adjustment cost, using equations (6), (7), and (9). These objects determine the residual short-term financing requirement through equation (11).

3. The maximum resources that can be raised through ex-post capital liquidation without violating the collateral constraint, as defined by equation (13). The support $[\phi_l, \phi_m]$ of the revenue shock is partitioned into twenty bins. For every state–choice pair and every bin, the algorithm records whether the realization leads to repayment without liquidation, repayment after liquidation, or default. When liquidation occurs, it also records the remaining capital stock $\tilde{k}'(\phi)$ implied by the liquidation condition.

4. The short-term debt price

$$p = H(s, y, k', b'),$$

determined by the creditors' zero-profit condition (20), together with the associated repayment, liquidation, and default probabilities and expected dividends.

When the residual financing requirement in equation (11) is nonpositive, the firm does not need short-term borrowing and the excess resources are distributed to shareholders as described in Section 3.5. When short-term borrowing is positive, the debt price is obtained from the creditors' break-even condition.

A state–choice pair is marked infeasible and is never selected if it violates the collateral constraint

$$b' \leq \lambda k'$$

or if the creditors' problem has no feasible short-term debt price. None of the objects described above depends on the value function and they therefore remain fixed during a discrete value-function iteration.

C.2 Why We Solve the Bellman Equation Directly

The model does not generate an Euler equation that can be conveniently inverted to recover the firm's current choices. A change in a production or financing choice generally affects several margins simultaneously: the current financing requirement, the endogenous short-term debt price, the repayment and default thresholds, the amount of capital liquidated following an adverse revenue realization, and the continuation value of the firm.

This interaction is especially transparent for short-term finance. The repayment factor is itself determined by the creditors' zero-profit condition in equation (20); hence a change in a firm's choices alters both the quantity of short-term finance and its price. The same choices also affect the default threshold in equation (18) and, through the liquidation value in equation (13), the transition between repayment, liquidation, and default states.

Capital and long-term debt introduce additional non-smoothness through partial irreversibility, the quadratic capital-adjustment cost, the absolute-value long-term-debt adjustment cost, the fixed long-term-debt participation cost, and the collateral constraint. The numerical solution therefore works directly with the Bellman problem in equation (24), allowing the maximizing choice to move across these constrained and regime-switching regions.

C.3 Discrete Value-Function Iteration

Let V^j denote the value function at iteration j. Conditional on a state (z, k, b) and a feasible choice (n, k', b'), short-term borrowing is determined from equation (11), and the short-term debt price is determined from equation (21).

Let

$$q = ps$$

denote the promised short-term repayment, and let ϕ^r and ϕ^e denote the repayment and default thresholds defined in equations (22) and (18). Given the objects precomputed above, the value of a feasible state–choice pair at iteration j is

$$\begin{aligned}\Gamma^j(z,k,b;n,k',b') = d + \beta\Bigg\{ & [1-G(\phi^r)]\,\mathbb{E}\left[V^j(z',k',b') \mid z\right] \\ & + \int_{\phi^e}^{\phi^r} \mathbb{E}\left[V^j\left(z',\tilde{k}'(\phi),b'\right) \mid z\right] g(\phi)\,d\phi \\ & + G(\phi^e)V^e(k',b',s)\Bigg\},\end{aligned} \tag{116}$$

where d is the expected current dividend implied by the policy described in Section 3.5. The expectation over z' is computed using the transition matrix associated with the discretized productivity process.

The integral over the liquidation region in equation (116) is evaluated using the twenty-bin partition of the revenue-shock distribution constructed during the precomputation stage. Each bin contributes the continuation value associated with its recorded post-liquidation capital stock $\tilde{k}'(\phi)$, weighted by the probability of that bin.

For every state, the Bellman update is

$$V^{j+1}(z,k,b) = \max_{n,k',b'} \Gamma^j(z,k,b;n,k',b'), \tag{117}$$

where the maximization is taken only over feasible choices. The location of the maximizing choice is stored as the discrete policy.

The iteration is initialized at

$$V^0 = 0$$

and proceeds until the supremum norm of the change in the value function falls below the prescribed convergence tolerance,

$$\left\|V^{j+1} - V^j\right\|_\infty < \varepsilon_V. \tag{118}$$

The discrete policy is stored as a sparse indicator selecting one choice column for each state.

C.4 Off-Grid Refinement

The discrete solution provides the starting point for a continuous refinement of the policy functions. States remain on the original state grid, but the choices (n,k',b') are allowed to lie between choice-grid nodes.

For each state, the converged choice-value array Γ is interpolated across the discrete choice grid. The implementation allows either linear interpolation or spline interpolation. The continuous objective is then maximized over the choice domain, starting from the discrete optimum and repeating the search from a short set of scale factors around that choice.

A candidate continuous choice is retained only if it satisfies the collateral constraint

$$b' \le \lambda k',$$

the creditors' break-even problem admits a feasible short-term debt price, and its interpolated value is at least

as large as the value of the discrete policy.

After all states have been refined, the continuation values associated with the liquidation region are rebuilt using the updated value function. The choice values in equation (116) and the maximization in equation (117) are then recomputed. This outer refinement loop is repeated until the same supremum-norm criterion falls below a tighter convergence tolerance.

For the construction of the transition matrix, each continuous policy (n, k', b') is placed on its surrounding grid nodes using multilinear weights determined by the distance from the continuous choice to those nodes. These weights replace the single nonzero indicator associated with the discrete policy. The off-grid policy and the stationary distribution generated by this weighted policy are used for the quantitative results.

C.5 Stationary Distribution

Once the policy functions have been obtained, they generate a transition law over firm states. The survival transition kernel is the discrete counterpart of $\mathcal{Q}$ in equation (26). For each current state, the transition probabilities combine the optimal policy, the productivity transition probabilities, and the probabilities of the three revenue-shock regions.

Conditional on survival without liquidation, the firm moves to the chosen capital and long-term debt position (k', b') and draws z' from the productivity Markov chain. Conditional on survival after liquidation, the firm moves to the capital stock $\tilde{k}'(\phi)$ associated with the relevant revenue-shock bin while retaining the chosen long-term debt position b'. Conditional on default and exit, the firm is replaced by a new entrant according to the entrant distribution η^e defined in equation (27).

Let $\mathbf{Q}$ denote the substochastic transition matrix associated with surviving firms before replacement. For state i, the probability of default and exit is

$$e_i = 1 - \sum_{j=1}^{N_x} Q_{ij}. \tag{119}$$

The transition matrix used to iterate the cross-sectional distribution, denoted by $\mathbf{T}$, adds entrant replacement to these survival transitions. Hence each row of $\mathbf{T}$ sums to one. Numerically, each row is checked against unity; any small discrepancy generated by the discretization of the revenue-shock distribution is removed by normalizing the row.

If $\boldsymbol{\mu}^{(j)}$ denotes the distribution of firms at iteration j, the implied mass of exits is

$$M^{e,(j)} = \boldsymbol{\mu}^{(j)}\mathbf{1} - \boldsymbol{\mu}^{(j)}\mathbf{Q}\mathbf{1}, \tag{120}$$

where $\mathbf{1}$ is a conformable vector of ones. Replacement entry implies that the same mass enters in the following period.

Using the row-stochastic transition matrix $\mathbf{T}$, the distribution is updated according to

$$\boldsymbol{\mu}^{(j+1)} = \boldsymbol{\mu}^{(j)}\mathbf{T}. \tag{121}$$

Starting from an initial distribution, iteration continues until

$$\left\| \boldsymbol{\mu}^{(j+1)} - \boldsymbol{\mu}^{(j)} \right\|_{\infty} < \varepsilon_{\mu}. \tag{122}$$

The resulting fixed point is the stationary distribution associated with the policy functions.

C.6 Equilibrium Wage

For a given wage w, the solution of the firm problem and the associated stationary distribution imply aggregate labor demand

$$L^d(w) = \sum_{x \in \mathcal{X}} n(x; w) \mu^*(x; w). \tag{123}$$

The stationary-equilibrium wage satisfies

$$L^d(w^*) = 1, \tag{124}$$

which is the discrete counterpart of equation (33).

Thus, for a given parameterization, the numerical steady state requires a wage for which the policy functions, the induced stationary distribution, and aggregate labor demand are mutually consistent.

C.7 Simulated Method of Moments

The parameters

$$\theta = (c_{ld}, c_f, c, \phi_l, \alpha, \gamma, \tau) \tag{125}$$

are calibrated jointly by Simulated Method of Moments. For every candidate parameter vector θ, the stationary equilibrium is solved and the same seven moments reported in Table 3 are constructed.

Let

$$m^D = \begin{bmatrix} 1.04 \\ 0.20 \\ 0.31 \\ 1.47 \\ 0.46 \\ 0.74 \\ 0.68 \end{bmatrix} \tag{126}$$

denote the data targets, corresponding respectively to

1. the average price of short-term relative to long-term debt;
2. the labor-to-sales ratio;
3. the capital-to-sales ratio;
4. the aggregate debt-to-capital ratio;
5. the average debt-to-sales ratio;
6. the share of entrants surviving for five years; and

7. the share of firms using both debt instruments.

Let $m^M(\theta)$ denote the corresponding vector of model moments. The calibration chooses

$$\widehat{\theta} = \arg\min_{\theta \in \Theta} \left[m^M(\theta) - m^D\right]' W \left[m^M(\theta) - m^D\right], \tag{127}$$

where W denotes the weighting matrix used in the calibration.

For each proposed θ, evaluating equation (127) requires solving the stationary equilibrium and constructing the corresponding model moments. The internally calibrated values reported in Table 2 are

$$\widehat{\theta} = (0.009,\ 0.041,\ 3.79,\ 0.149,\ 0.665,\ 0.186,\ 0.20)\,. \tag{128}$$

C.7.1 Construction of Calibration Moments

All model moments are constructed using definitions that correspond as closely as possible to their empirical counterparts. The five-year survival rate is computed from an entrant cohort rather than from the stationary replacement process. Let $\boldsymbol{\eta}^e$ denote the initial distribution of entrants and let $\mathbf{Q}$ denote the survival transition matrix evaluated at the stationary equilibrium. The fraction of the initial cohort surviving for five periods is

$$\text{Survival}_5 = \boldsymbol{\eta}^e \mathbf{Q}^5 \mathbf{1}. \tag{129}$$

Because $\mathbf{Q}$ is substochastic, firms that default disappear from this cohort rather than being replaced by new entrants.

The share of firms using both financing instruments is computed from the stationary distribution and the associated debt policies:

$$\text{BothDebt} = \frac{\sum_{x \in \mathcal{X}} \mathbf{1}\left\{s(x) > 0,\, b^+(x) > 0\right\} \mu^*(x)}{\sum_{x \in \mathcal{X}} \mu^*(x)}. \tag{130}$$

The remaining calibration moments are calculated from the stationary cross section using the corresponding firm policies and balance-sheet variables.

C.7.2 Weighting and Optimization

The calibration criterion in equation (127) allows the moments to differ in scale and precision through the weighting matrix W. Each evaluation of the criterion uses the same numerical solution procedure described above.

C.8 Computational Accuracy and Implementation Checks

Several checks are built directly into the numerical procedure. Choices that violate the collateral constraint or for which the creditors' break-even problem has no feasible solution are removed from the feasible set. During construction of the transition matrix, each row is checked to ensure that the probabilities sum to one, with any small discrepancy arising from the discretization of the revenue-shock distribution removed by normalization. The discrete value-function iteration, the off-grid refinement, and the stationary-distribution iteration are each

terminated using supremum-norm convergence criteria. The off-grid refinement accepts a continuous policy only if it is feasible and yields a value at least as high as the discrete policy from which the search is initialized.